Power System

Fundamentals of power System

→ Three basic elements of electrical engineering
- ↳ resistors → consume ohmic & dissipative energy
- ↳ Inductor → store in positive half cycle and give away in negative half cycle of the supply. The magnetic & electric field energies respectively.

Single phase transmission

Consider an inductive circuit.

Instantaneous Voltage $v = V_m \sin \omega t$

$\Rightarrow i = I_m \sin(\omega t - \phi)$ {Current lags voltage by ϕ in inductive}

Instantaneous power:

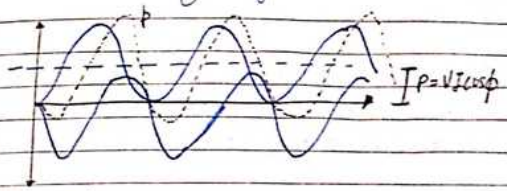
$$p = vi = V_m \sin \omega t \times I_m \sin(\omega t - \phi)$$
$$= V_m I_m \sin \omega t \sin(\omega t - \phi)$$
$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} (2 \sin \omega t \sin(\omega t - \phi))$$
$$= VI [\cos \phi - \cos(2\omega t - \phi)] \quad -(1)$$

(1) can be written as

$$p = VI \cos\phi (1 - \cos(2\omega t)) - VI \sin\phi \sin 2\omega t \quad -(2)$$
(I) (II)

(i) Due to component (I) in Eq(2) the waveform pulsates around the same average power $VI\cos\phi$ but never becomes negative as the factor $(1-\cos 2\omega t)$ can at most become zero. This is known as active power.

(ii) Due to component [II] in Eq(2) the waveform pulsates and has zero as its average value due to $\sin\phi$ which means it goes negative. This is known as reactive power and it does not do any useful work.



Eq(2) can be written as

$$p = P(1 - \cos 2\omega t) - Q \sin 2\omega t$$

P & Q both have the same dimensions of watts but Q is called the new active power and is measured in terms of Volt Amperes reactive (VAr)

Q requires more attention due to the properties of $\sin\phi$ which is -ve in case of capacitive circuits & +ve in the case of inductive circuits.

$\Rightarrow$ the capacitor is a generator of positive reactive VAr hence, it is better to consider a capacitor as providing a lagging current than as recieving a leading current.

The three phase transmission

Assuming that the system is balanced:

$V_a = V_m \sin \omega t$; $i_a = I_m \sin(\omega t - \phi)$
$V_b = V_m \sin(\omega t - 120°)$; $i_b = I_m \sin(\omega t - \phi - 120°)$
$V_c = V_m \sin(\omega t + 120°)$; $i_c = I_m \sin(\omega t - \phi + 120°)$

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The total power transmitted :-

$P = V_a i_a + V_b i_b + V_c i_c$

$= V_m \sin \omega t \, I_m \sin(\omega t - \phi) + V_m \sin(\omega t - 120°) \, I_m \sin(\omega t - \phi - 120°) + V_m \sin(\omega t + 120°) \, I_m \sin(\omega t - \phi + 120°)$

$= VI[2 \sin \omega t \sin(\omega t - \phi) + 2 \sin(\omega t - 120°) \sin(\omega t - \phi - 120°) + 2 \sin(\omega t + 120°) \sin(\omega t - \phi + 120°)]$

$= VI[\cos\phi - \cos(2\omega t - \phi) + \cos\phi - \cos(2\omega t - 240° - \phi) + \cos\phi - \cos(2\omega t + 240° - \phi)]$

$= VI[3\cos\phi - \cos(2\omega t - \phi) + \cos(2\omega t - \phi)]$

$\boxed{P = 3VI\cos\phi}$

Main Components of overhead lines

↳ Conductors : Carry electric power from the sending end station to recieving end station.
↳ Supports : Poles or towers which keep the conductors at a suitable distance from the ground.
↳ Insulators : Which are attached to supports and insulate the conductors from the ground.
↳ Cross arms : Provide support to the insulators
↳ Miscellaneous items : Phase plates, danger plates, lightening arrestors and anti-climbing wires.

Conductor Materials

The conductor material used should have the following properties :

(i) High electrical conductivity
(ii) High tensile strength in order to withstand mechanical stresses
(iii) Low cost so that it can be used for long distances.
(iv) Low specific gravity so that the weight per unit volume is small.

→ Most Commonly used Conductor Materials :
(i) Copper
(ii) Aluminium
(iii) Steel cored aluminium
(iv) Galvanised steel
(v) Cadmium Copper

If there are n layers in a stranded conductor wire, the total number of individual wire is $[3n(n+1)+1]$ & diameter is $(1+2n)d$.

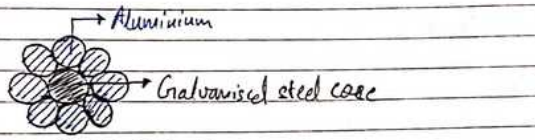
1. Copper
   ↳ Ideal material for overhead lines
      ↳ high electrical conductivity
      ↳ Great tensile strength
   ↳ Always used in hard drawn form in stranded conductor
      ↳ Decreases electrical conductivity, slightly.
      ↳ Increases tensile strength considerably.
   ↳ High Current density
      ↳ Lower Cross sectional area possible
      ↳ Lower Area means lower area for wind loads.
   ↳ High cost

2. Aluminium
   ↳ Cheaper and lighter compared to Cu
      ↳ Lower conductivity

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↳ lower tensile strength
↳ For the same resistance the diameter of Al required is 1.26 times Cu
↳ Greater wind pressure => higher towers and greater sag
↳ Specific gravity lower than copper
↳ supports needn't be as strong
↳ liable to swings => larger cross arms required.

3. Steel Cored Aluminium



↳ Greater tensile strength
↳ Longer distances because of less sag
↳ Towers of smaller height can be used.

4. Galvanised Steel

↳ Very high tensile strength
↳ Can be used for extremely long distances
↳ Can be used for short distances where conditions are harsh.

5. Cadmium Copper

↳ Addition of 1% or 2% of cadmium to Cu increases tensile strength by 50%.
↳ Conductivity only reduces by 15%.

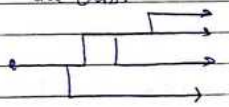
Configuration of transmission lines

1. Single line : Power is supplied from the generation station to the load centre through a single line ckt.

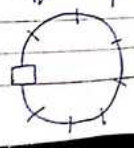
2. Parallel lines : Two ckts are placed parallel either on the same supports or separate supports. This is done so that if one breaks, the other would still transmit electricity usually used when bulk power is transmitted.



3. Radial lines : Power plant or substation provides electricity to the neighbouring territory via radial lines which may consist of parallel & ckt single line ckts.



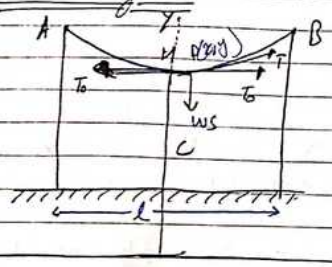
4. Ring System : For systems covering a large territory we use ring system. The main high voltage power lines makes a ring with various tappings at different points. The ring is supplied power from both sides.



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5. Networks: Several interlocking ring/loop systems form a network.

The Catenary Curve



T_0 = tension at lowest point 'A' (in kg)
Consider the equilibrium of the small length s of the wire up to points P(x,y). The 3 forces acting on the wire are:-

(i) The horizontal tension $T_0 = wc$
(ii) The vertical

Sag in Overhead lines
in level

The difference between the points of the supports and the lowest point on the conductor is sag.

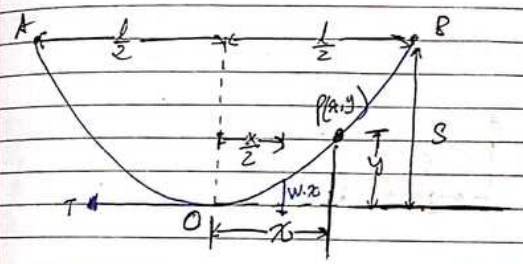
→ When a conductor is ~~fixed~~ suspended between two supports at the same level, it takes the shape of a catenary. However, if the sag is very small compared to the span, then the sag-span curve is like a parabola.

Calculation of Sag

→ Sag should be adjusted such that the tension is at acceptable level considering the appropriate ground clearance.

→ It is standard practice to keep the conductor tension about 50% of its ultimate tensile strength. i.e. the safety factor with respect to the conductor tension should be at minimum 2.

(i) When supports are at equal levels



Let, l = length of span
w = weight per unit length of the conductor
T = Tension in the conductor.

Consider a point P(x,y) and assuming the catenary is very small
$\Rightarrow$ (OP = x)
The two forces acting on the portion OP
(a) The weight wx of the conductor acting at point $\frac{x}{2}$ from O.
(b) The tension T acting at O

Equating the moments of the two forces about point O:

$$Ty = (wx) \times \left(\frac{x}{2}\right)$$

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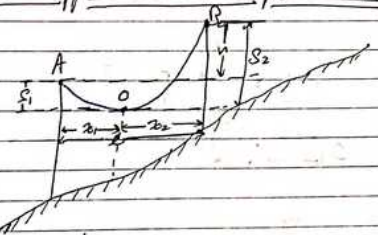
$$y = \frac{wx^2}{2T}$$

For maximum dip (sag) y

$x = l/2$ & $y = s$

$$\text{Sag, } s = \frac{wl^2}{8T}$$

(ii) When supports are at unequal level



Let, l = span length

h = Difference in levels between two supports

x_1 = Distance of support from A [i.e. lower level]

x_2 = Distance of O from B [i.e. higher level]

T = Tension in the conductor

If w is the weight per unit length of the conductor

$$\text{Sag } s_1 = \frac{wx_1^2}{2T}$$

$$\text{Sag } s_2 = \frac{wx_2^2}{2T}$$

$$x_1 + x_2 = l \quad -(1)$$

$$s_2 - s_1 = \frac{w}{2T}[x_2^2 - x_1^2] = \frac{w}{2T}[x_2 + x_1][x_2 - x_1]$$

$$s_2 - s_1 = \frac{wl}{2T}(x_2 - x_1)$$

$$\Rightarrow x_2 - x_1 = \frac{2Th}{wl} \quad [\because s_2 - s_1 = h] \quad -(2)$$

Solving (1) & (2)

$$x_1 = \frac{l}{2} - \frac{Th}{wl}$$

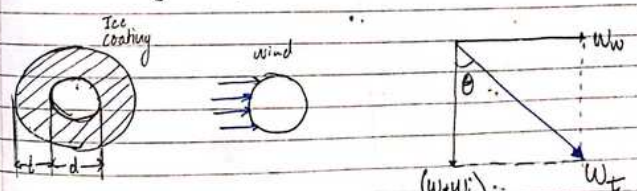
$$x_2 = \frac{l}{2} + \frac{Th}{wl}$$

Effect of wind and ice loading

→ The weight of ice acts vertically downwards i.e., in the same direction as the weight of the conductor.

→ The force due to wind is assumed to act horizontally i.e., at right angle to the projected surface of the conducta.

∴ The total force on the conductor is a vector sum of the vertical & horizontal forces.



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Total weight of the conductor per unit length is,

$W_c = \sqrt{(w+w_i)^2 + (W_w)^2}$

Where, w = weight of conductor per unit length
= Conductor material density × volume per unit length

w_i = weight of ice per unit length
= density of ice × volume of ice per unit length

= density of ice $\times \frac{\pi}{4}[(d+2t)^2 - d^2] \times 1$

= density of ice $\times \pi t(d+t)$

W_w = Wind force per unit length
= wind pressure per unit area × projected area per unit length.

= wind pressure $\times [(d+2t) \times 1]$

When both wind & ice loading act on a conductor :-

(ii) The conductor sets itself in a plane at an angle θ to the vertical when

$$\tan\theta = \frac{W_w}{w + w_i}$$

(iii) The sag in the conductor is given by :

$$S = \frac{W_c l^2}{8T}$$

S represents the stand sag in a direction making an angle θ with the vertical

(iv) The vertical sag = $S\cos\theta$

Some Mechanical principles

(i) Tower height

- Depends upon the length of span
  - Longer span means fewer towers but must be tall and costly
  - Sagging hazards increase greatly with increase in height
    - Horizontal spacing is preferred even though wider right of way is required.

(ii) Conductor Clearance to Ground

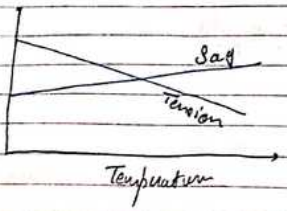
- Clearance at the time of greatest sag should be around 6 to 12 m

(iii) Sag and tension

- When laying overhead transmission lines, necessary to allow reasonable factor of safety in respect to the tension to which the conductor is subjected
  - Relationship between sag and tension depends upon the loading conditions
    - For ex: tension increases when temp. decreases because decrease in Sag
  - In planning sag, tension & clearance, a maximum stress is selected which is to be developed at the worst possible conditions.
- Wind loading increases the sag in the resultant loading direction but decreases in the vertical component.
  - Wind loading is not considered in clearance calculation
    - Unless vertical clearance necessary.

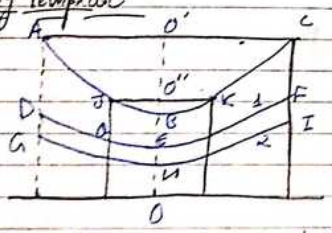
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(iv) Stringing Charts



These charts are very helpful while stringing overhead lines.

(v) Sag Template



↳ Two types of supports are used. ① Standard (straight run/intermediate) ② Angle/anchor/tension

↳ Standard towers are used for straight runs & normal conditions.
↳ Angle towers are used at angles, terminals and other points where there is a considerable amount of unbalanced pull on the supports.
↳ Angle towers are designed to withstand heavy loading as compared to standard towers.

↳ In order to locate the position of towers
↳ First step: Known suitable value of support height.
↳ Second step: To plot a sag template on transparent paper.

→ The horizontal & vertical distances are the span lengths & sag respectively.
→ GHI is tower footing line [location of footing of the towers]
→ DEF is ground clearance line [Minimum clearance of the conductor]
→ Curve ABC → span length AC
↳ sag O'B
→ Curve JBK → span length JK
↳ sag O''B

(vi) Equivalent Spans

Equivalent span length, $l = \sqrt{\frac{l_1^3 l_1 + l_2^3 + \dots}{l_1 + l_2 + l_3 + \dots}}$

l_1, l_2, l_3 → lengths of individual spans b/w two tension towers.

(vii) Conductor Vibrations

↳ If wind velocity is small, swinging of conductors is harmless
↳ provided clearance is sufficient, large
↳ conductors don't in spacing distance

↳ Overhead transmission lines experience vibrations in the vertical plane and are of two types :- ① Aeolian vibrations
② Galloping or dancing of conductors

→ Aeolian vibrations :- → High frequency, low amplitude
↳ frequency of the order of 5 to 40 Hz (cycle per second)
↳ Amplitude of the order of 2cm to 30cm
↳ Span length of 1m to 10m
↳ Wind velocity about 4 to 12 km/hr & steady and uniform

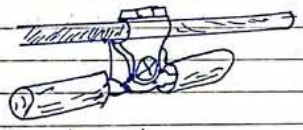
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↳ At a heavy anchor clamp, a travelling wave will be reflected with a momentary bending of the cable about the width of the clamp
↳ A light anchor clamp, very freely supported will reduce the bending

↳ Besides design of clamps, vibration are reduced usually :-
↳ (a) Reinforcement or armour rods
(b) Dampers

↳ (a) → Vatercy System of Reinforcement
↳ Conductor at the suspension clamp is surrounded by a spiral layer of small round rods, tapered at each end and larger in diameter as compared to individual wires.
↳ Armoured rods also provide protection against flashover.

↳ (b) → True dampers → special device designed to absorb vibrational energy
↳ Stock-bridge damper



↳ Extremely simple
↳ Effective for suppressing high frequency vibrations
↳ The weights are of galvanized pipes or zinc filled with lead
↳ It is designed to be tuned either when line is live or dead
↳ Two dampers are required, one at either point of suspension
↳ The damping action is due to dissipation of vibrational energy by hysteresis and inter strand friction in flexible damper cable.

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→ Galloping or dancing of conductors :→ low frequency & high amplitude
↳ Frequency of the order of $\frac{1}{4}$ to $1\frac{1}{2}$ cycles/sec and amplitude of about 6 meters
↳ These are self excited type vibrations
↳ When wind blows past a non-circular conductor
↳ Builds with itself
↳ results in flexure fatigue & flashover
↳ To damp these oscillations it is required to make the conductor circular
↳ For stranded cond. → PVC tape is wrapped
↳ If sleet/ice coating present → increasing I²R losses but not studied.

Overhead line insulators

↳ Provide insulation to the power conductor from the ground.
↳ Mainly made up of glazed porcelain or toughened glass.
↳ Porcelain → (i) Silica 20%
(ii) Feldspar 30%
(iii) Clay 50%
↳ Porcelain should be ivory white, smooth and free from defects
↳ Vitrified because pores and air inside reduces the dielectric strength of porcelain
↳ Dielectric strength should be 15KV to 17KV for every one tenth inch thick
↳ Toughened glass has very high dielectric strength (35KV for one tenth inch thickness)
↳ transparent (Easy to see defects & flaws)
↳ Lower coefficient of thermal expansion
↳ Drawback : Moisture condenses very easily
↳ Use limited to about 22KV.
↳ To avoid fractures from expansion and contraction, cement is used b/w metal and porcelain

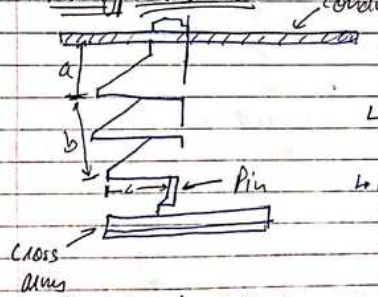
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Types of Insulators

↳ 3 types of insulators used in Overhead lines:-
(i) Pin Type
(ii) Suspension Type
(iii) Strain Type

↳ Desirable properties: (i) High mechanical Strength to withstand conductor, wind etc.
(ii) High electrical resistance to avoid leakage current to earth.
(iii) High relative permittivity so, that dielectric strength is high.
(iv) Non-porous, free from impurities and cracks or the permittivity will be lowered.
(v) High ratio of puncture strength to flashover.

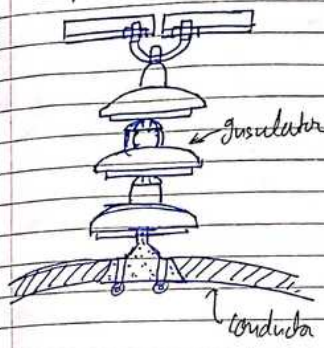
↳ Pin type Insulators



↳ Secured to cross arms on the poles
↳ There is a groove on the upper end of the insulator
↳ One conductor passes through the groove bounded by the annealed wire of the same material

↳ Pin type insulators are used for voltages upto 33kV.
↳ Beyond 33kV, becomes too bulky and hence uneconomical

↳ Suspension Type Insulators



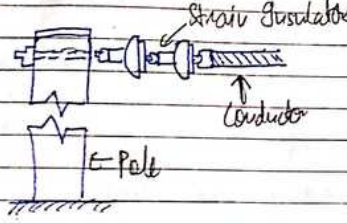
↳ Cost of pin type increases rapidly with operating voltage.
↳ For voltages > 33kV, suspension type is used
↳ Conductor is suspended at the bottom end while the other end is connected to the crossarm
↳ Each disc is about 11kV
↳ For 66kV, we would use 6 discs.

↳ Advantages of Suspension over pin type:
(i) Suspension type are cheaper than pin type for voltages > 33kV
(ii) Each disc is designed for low voltages, desired no. of discs can be used.
(iii) If one disc is damaged, the whole string isn't useless
(iv) Suspension arrangement provides greater flexibility, the insulator string can swing in any direction and take the position where mechanical stresses are minimum.
(v) In case of increased demand, it is always easier to increase line voltage by increasing no. of conductors, than the insulation required for the increased voltage can be easily obtained from by adding additional discs.
(vi) Suspension type are usually used with steel towers, as the conductors are suspended below the crossarms which are earthed, they provide partial lightening protection as well.

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Strain Insulators

↳ At dead ends, corners or sharp curves, there is great tension on the lines.
  ↳ Strain insulators are used to relieve the excessive tension.
↳ When tension is exceedingly high like in long river spans, two or more strings are used in parallel.



↳ Discuss Causes of insulator failure :
  ↳ Insulators are required to withstand mechanical & electrical stresses
  ↳ Electric breakdown of an insulator can occur either by flashover or puncture.
  ↳ In flashover, an arc occurs between the live conductor and insulator pin (i.e, earth)
    ↳ The discharge jumps across the air gap, following shortest distance.
  ↳ In case of flashover, insulator continues to operate if not destroyed
  ↳ In puncture, the discharge b/w conductor and pin occurs through the body of the insulator.
    ↳ The insulator gets permanently destroyed
    ↳ Sufficient thickness is provided to prevent this.

↳ Ratio of puncture strength to flashover voltage is known as safety factor :

$$\text{Safety factor of insulator} = \frac{\text{Puncture Strength}}{\text{Flash-over voltage}}$$

Potential distribution over Suspension Insulator String

↳ Each disc in suspension insulator string forms a capacitor C.
  ↳ This is known as mutual or Self-Capacitance
↳ A capacitance also exists between the discs (metal fittings) & the tower (earth)
  ↳ This is known as Shunt Capacitance C_1.
    ↳ Due to this charging current is not the same through all the discs
      ↳ Voltage across each disc is different
        $(V_4 > V_3 > V_2 > V_1)$



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String Efficiency

↳ The disc nearest to the conductor has much higher potential than other discs
  ↳ Under maximum electrical stress ⇒ likely to be punctured
    ↳ Means should be provided to equalize the potential across each unit.

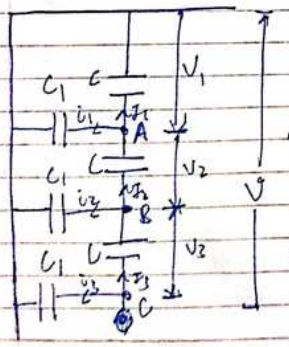
↳ This unequal potential distribution is undesirable and is usually expressed in terms of string efficiency.

$$\text{String efficiency} = \frac{\text{Voltage across the string}}{n \times \text{Voltage across disc nearest to conductor}}$$

n = number of discs

↳ Greater the string efficiency, the more uniform is the voltage distribution.

↳ Mathematical Expression:



Let us assume self capacitance of each disc is C & mutual C_1
and, $C_1 = kC$
Applying KCL at A,

$I_2 = i_1 + I_1$

$V_2\omega C = V_1\omega C + V_1\omega C_1$

$\Rightarrow V_2\omega C = V_1\omega C + V_1\omega kC$

$\boxed{V_2 = V_1(1+k)}$ — (1)

Applying KCL at B,

$I_3 = I_2 + i_2$

$V_3\omega C = V_2\omega C + (V_1+V_2)\omega C_1$

$V_3\omega C = V_2\omega C + (V_1+V_2)\omega kC$

$V_3 = V_2 + (V_1+V_2)k$

$= kV_1 + V_1(1+k)$

$V_3 = kV_1 + V_1(1+k)^2$

$V_3 = V_1[k + (1+k)^2]$

$\boxed{V_3 = V_1[1+3k+k^2]}$ — (2)

$\because V = V_1 + V_2 + V_3$

$\Rightarrow V = V_1 + V_1(1+k) + V_1[1+3k+k^2]$

$= V_1(3+4k+k^2)$

$\boxed{V = V_1(1+k)(3+k)}$ — (3)

From (1), (2) & (3)

$$\frac{V_1}{1} = \frac{V_2}{(1+k)} = \frac{V_3}{1+3k+k^2} = \frac{V}{(1+k)(3+k)}$$

Voltage drop across top unit: $\boxed{V_1 = \frac{V}{(1+k)(3+k)}}$

↳ The greater the value of k, the more non-uniform is the potential across the discs
  ↳ Lesser String efficiency

↳ Inequality in voltage distribution increases with increase in discs
  ↳ Shorter strings are more efficient.

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Methods of Improving String Efficiency

↳ (i) By using longer cross arms

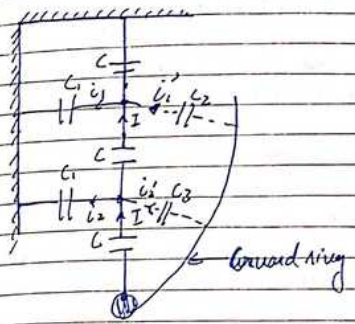
↳ String efficiency depends upon k i.e. C_1/C
↳ Greater the k, greater the efficiency
↳ Can be increased by reducing shunt capacitance C_1
↳ Distance of conductor from tower should be increased
↳ Due to cost and strength limitations, k = 0.1 is the limit that can be achieved by this method.

↳ (ii) By grading the insulators

↳ Insulators of different dimensions are chosen
↳ Each has a different capacitance.
↳ Erected in a way that top unit has the minimum capacitance, increasing progressively as the bottom unit is reached
↳ Since voltage is inversely proportional to capacitance
↳ This method equalises the potential distribution

↳ (iii) By using a guard ring

↳ Potential across each unit in a string can be equalised by using a guard ring
↳ Metal ring electrically connected to the conductor and surrounding the bottom insulator.
↳ It induces a capacitance b/w metal fittings and line conductor.
↳ Contoured in such a way that shunt capacitance currents are equal to the metal fitting line capacitance currents.
↳ Same charging current I flows through each unit
↳ Potential distribution is uniform



↳ Points to remember while solving string efficiency problems:

(i) Max. voltage appears across disc nearest to the conductor
(ii) Voltage across the string = phase voltage = Voltage b/w line and earth
(iii) line voltage = $\sqrt{3}$ × Voltage across string.

Corona

↳ When an alternating potential is applied across two conductors whose spacing is large as compared to their diameters, there is no change in the condition of atmospheric air surrounding them if the applied voltage is low.
↳ If the applied voltage exceeds a certain value called, critical disruptive voltage
↳ The conductors are surrounded by a faint violet glow called corona.
↳ The phenomena of corona is accompanied by a hissing sound, production of ozone, power loss & radio interference.
↳ Higher the voltage, the greater are the above attributes.
↳ If the voltage increased to breakdown value
↳ flashover will occur b/w the conductors

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↳ The phenomenon of violet glow, hissing sound and production of ozone gas in an overhead transmission line when the applied voltage exceeds the critical disruptive voltage is known as Corona.

↳ If the conductors are polished and smooth
↳ glow will be uniform
↳ If not, the rough points will appear brighter.
↳ If on D.C., the +ve wire has uniform glow but -ve wire has spotty glow

Theory of Corona formation

↳ Some ionization always present in air due to cosmic rays, ultra violet radiations and radioactivity
↳ Under normal conditions, air around conductors contains ions/free e⁻
↳ When P.d applied b/w conductors, potential gradient is set up which has max. value at the surfaces of the conductors.
↳ Under the influence of the potential gradient, existing free electrons acquire greater energy (velocities)
↳ Greater the applied voltage, greater is the velocity of free e⁻.
↳ When the potential gradient at the conductor surface reaches 30 kV/cm per cm (max value)
↳ Velocity of free e⁻ sufficient to dislodge more e⁻ from neutral molecules
↳ produces another ionized free e⁻ which is driven an accelerated until they collide with other neutral molecules.
↳ This process of ionization is cumulative and results in either corona or spark b/w the conductors.

Factors affecting Corona

↳ Corona is affected by the physical state of the atmosphere as well as the conditions of the line.

(i) Atmosphere: ↳ On stormy weather, the number of ions is more than normal and as such corona takes place at much less voltage as compared to fair weather.

(ii) Conductor size: ↳ Rough and irregular surface will give rise to more corona
↳ Unevenness of surface reduces the value of breakdown voltage.
↳ ∴ A stranded conductor gives rise to more corona than a solid conductor.

(iii) Spacing between Conductors: ↳ If the spacing b/w conductors is made very large as compared to their diameter, there may not be any corona effect.
↳ Larger distance means reduced electrostatic stresses at the conductor surface.

(iv) Line Voltage: ↳ If it is low, there is no change in the surrounding air
↳ If value such that electrostatic stresses developed at cond. surface and value surrounding air conductive, corona is formed.

Terms used in the analysis of corona effect

(i) Critical disruptive voltage:
↳ the minimum phase-neutral voltage at which corona occurs.

↳ Consider two conductors of radii r cm and spaced d cm apart

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V is the phase-neutral potential,

Potential gradient at the conductor surface is given by,

$$g = \frac{V}{r \log_e \frac{d}{r}} \text{ V/cm}$$

↳ In order for corona to occur, g = breakdown strength of air
↳ breakdown strength of air at 76cm pressure and temp 25°C is 30kV
↳ 21.2 kV/cm (r.m.s) and denoted by g_0
↳ V_c is the phase-neutral voltage required under these conditions

$$\Rightarrow g_0 = \frac{V_c}{r \ln \frac{d}{r}}$$

where g_0 = 30kV/cm (max) or 21.2kV/cm (r.m.s)

⇒ Critical disruptive voltage $V_c = g_0 r \ln \frac{d}{r}$

↳ Above expression is for standard conditions
↳ Breakdown strength at pressure b and temperature t becomes δg_0, where δ

$$\delta = \text{air density factor} = \frac{3.92\,b}{273 + t}$$

∴ Critical disruptive voltage, $V_c = g_0 \delta r \ln \frac{d}{r}$

But, corrections for surface irregularity must also be made.
↳ multiplying by irregularity factor m_0.

∴ Critical disruptive voltage, $V_c = m_0 g_0 \delta r \ln \frac{d}{r}$ kV/phase

↳ where, m_0 = 1 [Polished conductors]
= 0.98 to 0.92 [Dirty conductors]
= 0.87 to 0.8 [Stranded conductors]

(ii) Visual Critical Voltage:
↳ It is the minimum phase-neutral voltage at which corona glow appears all along the line conductors.

↳ In parallel conductors, corona glow does not occur at critical disruptive voltage V_c, but at a higher voltage called Visual Critical voltage V_v.

$$V_v = m_0 g_0 \delta r \left(1 + \frac{0.3}{\sqrt{\delta r}}\right) \ln \frac{d}{r} \text{ kV/phase}$$

(iii) Power loss due to Corona:
↳ formation of corona is always accompanied by energy loss in the form of heat, light, sound and chemical action.

$$P = 242.2 \left(\frac{f + 25}{\delta}\right) \sqrt{\frac{r}{d}} (V - V_c)^2 \times 10^{-5} \text{ kW/km/phase}$$

where,
f = supply frequency in Hz
V = phase-neutral voltage (r.m.s)
V_c = disruptive voltage (r.m.s) per phase

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Advantages and Disadvantages of Corona

↳ For correct design of high voltage overhead lines, a balance must be struck b/w the advantages and disadvantages.

Advantages:

↳(i) Due to corona formation, the virtual diameter of the conductors is increased which reduces the electrostatic stresses b/w the conductors.

↳(ii) Corona reduces the effects of transients produced by surges.

Disadvantages:

↳(i) Corona is accompanied by a loss of energy
↳ affects transmission efficiency of the line.

↳(ii) Ozone is produced by corona
↳ Corrosion due to chemical action.

↳(iii) The current drawn by the line due to corona is non-sinusoidal
↳ Non-sinusoidal voltage drop occurs in the line
↳ Causes inductive interference with neighbouring communication lines.

Methods of Reducing Corona Effect

↳ Intense corona effects are observed at a working voltage (33kV etc.)
↳ Corona must be avoided or flashover can occur b/w the insulators or towers

(i) By increasing the conductor size:
↳ Voltage at which corona occurs is raised
↳ ACSR which have larger cross-sectional area are used in OHTL

(ii) By increasing conductor spacing:
↳ Voltage at which corona occurs is raised
↳ spacing can only be increased to a limit due to cost & strength limitation.

Resistance of Transmission lines

↳ Resistance of transmission line is the most important cause of power loss.

$$R = \frac{\rho l}{A}$$

↳ Variation resistance with temp:

$$R_2 = R_1[1 + \alpha_1(t_2 - t_1)]$$

where,

$$\alpha_1 = \frac{\alpha_0}{1 + \alpha_0 t_1}$$

α_0 = temperature coefficient at 0°C.

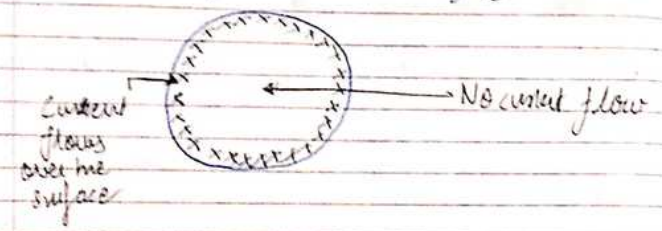
↳ In a 1-ϕ 2-wire DC line, the total resistance (loop resistance) is double the resistance of either conductor.

↳ In case of 3ϕ transmission line, resistance per phase is the resistance of one conductor.

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Skin Effect

↳ When DC current flowing
↳ Current uniformly distributed over the whole X section
↳ When AC current flowing
↳ Tendency to concentrate near the surface of the conductor
↳ Skin Effect.
↳ Due to this, the effective area of X section is reduced
↳ Resistance of the conductor is slightly increased.



↳ Solid conductor can be considered as having a large no. of strands
↳ The inductance of each strand will vary according to its position
↳ Strands near the centre are surrounded by greater magnetic flux
↳ Higher inductance → higher resistance
↳ Causes current to flow near the surface

↳ Skin effect depends upon :-
(i) Nature of material
(ii) Diameter of wire [increases with diameter]
(iii) Frequency [increases with increase in frequency]
(iv) Shape of wire [less for stranded as compared to solid]

Proximity Effect

↳ The alternating flux in a conductor caused by the current flowing in a neighbouring conductor gives rise to a circulating current which causes an apparent increase in the resistance of the conductors.

Flux Linkages

↳ Inductance of a circuit is the flux linkage per unit current

↳ Flux linkages due to a single current carrying conductor :
↳ Consider a long straight cylindrical conductor of radius r and current I A (rms)
↳ Magnetic field will exist inside the conductor as well as outside

(i) Flux linkage due to internal flux :
↳ Magnetic field intensity at a point x m from the centre is given by,

$$H_x = \frac{I_x}{2\pi x}$$



Assuming uniform current density,

$$I_x = \frac{\pi x^2 I}{\pi r^2} = \frac{x^2 I}{r^2}$$

$$H_x = \frac{x^2}{r^2} I_x \frac{1}{2\pi x} = \boxed{\frac{xI}{2\pi r^2} \text{ AT/m}}$$

if $\mu (= \mu_0 \mu_r)$ is the permeability of the conductor, flux density,

$$B_x = \mu_0 \mu_r H_x \text{ wb/m}^2$$

$$= \frac{\mu_0 \mu_r x I}{2\pi r^2} = \boxed{\frac{\mu_0 x I}{2\pi r^2} \text{ wb/m}^2}$$ [$\mu_r = 1$ for non-magnetic material]

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flux $d\phi$ through cylindrical shell of radial thickness dx and axial length 1 m,

$$d\phi = B_x \times 1 \times dx = \boxed{\frac{\mu_0 I x}{2\pi x^2}\, dx \text{ wb}}$$

⇒ flux linkages per meter length of the conductor

$$d\Psi = \frac{\pi x^2}{\pi x^2}\, d\phi$$

$$d\Psi = \frac{\mu_0 I x^3}{2\pi x^4}\, dx \text{ wb-turns}$$

⇒ Total internal flux linkages :-

$$\Psi = \int_0^x \frac{\mu_0 I x^3}{2\pi x^4}\, dx$$

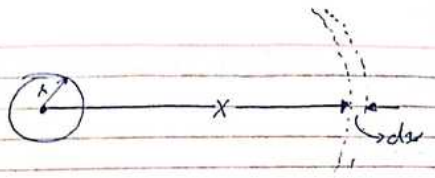
$$= \frac{\mu_0 I}{2\pi x^4} \int_0^x x^3 dx$$

$$= \frac{\mu_0 I}{2\pi x^4} \left[\frac{x^4}{4}\right]$$

$$\boxed{\Psi_{int} = \frac{\mu_0 I}{8\pi} \text{ wb-turns/meter length}}$$

(iii) Flux linkages due to external flux :

↳ External flux extends from the surface of the conductor to infinity.



↳ Field intensity at a distance x meters outside the conductor:

$$H_x = \frac{I}{2\pi x} \text{ AT/m}$$

Flux density, $B_x = \mu_0 H_x = \frac{\mu_0 I}{2\pi x}$ wb/m²

Flux

$$d\phi = B_x \times 1 \times dx = \frac{\mu_0 I}{2\pi x}\, dx \text{ wb}$$

∴ Flux linkages, $d\Psi = d\phi = \frac{\mu_0 I}{2\pi x}\, dx$ (as it links all the current in the conductor)

$$\boxed{\Psi_{ext} = \int_x^\infty \frac{\mu_0 I}{2\pi x}\, dx \text{ wb-turn}}$$

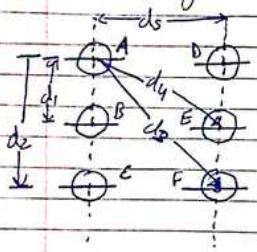
Overall flux linkages, $\Psi = \Psi_{int} + \Psi_{ext}$

$$= \frac{\mu_0 I}{8\pi} + \int_x^\infty \frac{\mu_0 I}{2\pi x}\, dx$$

$$\boxed{\Psi = \frac{\mu_0 I}{2\pi} \left[\frac{1}{4} + \int_x^\infty \frac{dx}{x}\right] \text{ wb-turns/m length}}$$

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↳ Flux linkages in parallel current carrying conductors



Flux linkages in A due to its own current,

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \int_{r_1}^{\infty}\frac{dx}{x}\right] \quad -(i)$$

Flux linkages in A due to I_B

$$= \frac{\mu_0 I_B}{2\pi}\int_{d_1}^{\infty}\frac{dx}{x} \quad -(ii)$$

Flux linkages in A due to I_C

$$= \frac{\mu_0 I_C}{2\pi}\int_{d_2}^{\infty}\frac{dx}{x} \quad -(iii)$$

Total flux linkage in A,

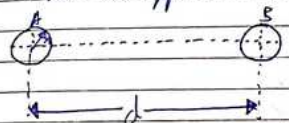
$$= (i) + (ii) + (iii) + \dots$$

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \int_{r_1}^{\infty}\frac{dx}{x}\right] + \frac{\mu_0 I_B}{2\pi}\int_{d_1}^{\infty}\frac{dx}{x} + \frac{\mu_0 I_C}{2\pi}\int_{d_2}^{\infty}\frac{dx}{x} + \dots\dots$$

↳ Similarly flux linkages with other conductors can be determined.

Inductance of a single phase two wire line

↳ A single phase line consists of two parallel conductors forming a loop of one turn.
↳ AC current sets up a changing magnetic flux
↳ links the loop and hence, it possesses inductance.
↳ As the X-sectional area of the loop is very large
↳ Even for small flux density, the total flux linking the loop is quite large
↳ Line has appreciable inductance.



↳ $I_A = I_B$ but in opposite directions because core forms the return circuit of the other.

$\therefore I_A + I_B = 0$

Flux linkages with conductor A

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \int_{r_1}^{d}\frac{dx}{x}\right] + \frac{\mu_0 I_B}{2\pi}\int_{d}^{\infty}\frac{dx}{x}$$

$$= \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} + \int_{r_1}^{\infty}\frac{dx}{x}\right) I_A + \left(\int_{d}^{\infty}\frac{dx}{x}\right) I_B\right]$$

$$= \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} + \ln\infty - \ln r_1\right) I_A + \left(\ln\infty - \ln d\right) I_B\right]$$

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$$= \frac{\mu_0}{2\pi}\left[\frac{I_A}{4} + \ln\infty\,(I_A + I_B) - I_A \ln r - I_B \ln d\right]$$

$$= \frac{\mu_0}{2\pi}\left[\frac{I_A}{4} - I_A \ln r - I_B \ln d\right]$$

$$= \frac{\mu_0}{2\pi}\left[\frac{I_A}{4} - I_A \ln r + I_A \ln d\right]$$

$$= \frac{\mu_0}{2\pi}\left[I_A\left(\frac{1}{4} + \ln\frac{d}{r}\right)\right]$$

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \ln\frac{d}{r}\right] \text{ wb-turn/m}$$

Inductance, $L_A = \frac{\Psi_A}{I_A}$.

$$= \frac{\mu_0}{2\pi}\left[\frac{1}{4} + \ln\frac{d}{r}\right] \text{H/m}$$

$$= 2\times10^{-7}\left[\frac{1}{4} + \ln\frac{d}{r}\right] \text{H/m}$$

$$= 2\times10^{-7}\left[\ln e^{1/4} + \ln\frac{d}{r}\right] \text{H/m}$$

$$= 2\times10^{-7}\left[\ln\frac{de^{1/4}}{r}\right] \text{H/m}$$

$$= 2\times10^{-7}\left[\ln\frac{d}{re^{-1/4}}\right] \text{H/m}$$

Let $r' = re^{-1/4}$

$$= 2\times10^{-7}\ln\frac{d}{r'} \text{ H/m}$$

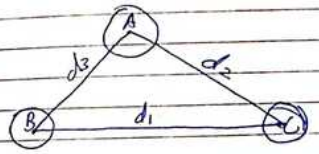
↳ r' is that of a fictitious conductor assumed to have no internal flux but with the same inductance as the actual conductor.

$e^{-1/4} = 0.7788$

$\Rightarrow re^{-1/4} = r' = 0.7788r$

↳ the term r' is called geometric mean radius (GMR)

Inductance of a 3 phase Overhead line



↳ A, B, C carrying current I_A, I_B, I_C resp.

$I_A + I_B + I_C = 0$

Total flux linkages in A,

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \int_r^\infty \frac{dx}{x}\right] + \frac{\mu_0 I_B}{2\pi}\int_{d_3}^\infty \frac{dx}{x} + \frac{\mu_0 I_C}{2\pi}\int_{d_2}^\infty \frac{dx}{x}$$

$$= \frac{\mu_0 I_A}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I_A - I_B \ln d_3 - I_C \ln d_2 + \ln\infty\,(I_A + I_B + I_C)\right]$$

$$\Psi_A = \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I_A - I_B \ln d_3 - I_C \ln d_2\right]$$

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(i) Symmetrical Spacing

↳ A, B, C are placed symmetrically at the corners of equilateral Δ of side d.
↳ $d_1 = d_2 = d_3 = d$

$$\psi_A = \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I_A - I_B \ln d - I_C \ln d\right]$$

$$\psi_A = \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I_A - (I_B + I_C) \ln d\right] \qquad (I_B + I_C = -I_A)$$

$$= \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I_A + I_A \ln d\right]$$

$$= \frac{\mu_0 I_A}{2\pi}\left[\frac{1}{4} + \ln\frac{d}{r}\right] \text{ wb-turns/m}$$

Inductance, $L_A = \frac{\mu_0}{2\pi}\left[\ln\frac{d}{r'}\right]$

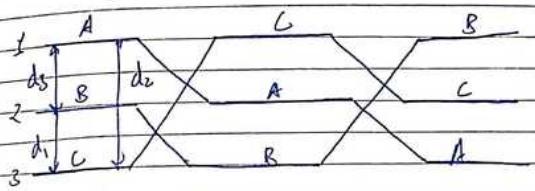
$$= 2\times10^{-7} \ln\frac{d}{r'} \text{ H/m}$$

(ii) Unsymmetrical Spacing

↳ Not equidistant
↳ Flux linkages hence, inductance of each phase different
↳ Unequal voltage drop in each phase even if current balanced
↳ Receiving end voltage not same for all phases

↳ Inorder to have equal voltage drops in all the conductors
↳ positions of the conductors at regular intervals along the line interchanged

↳ each conductor occupies the original position of every other conductor over an equal distance.
↳ known as transposition.



↳ the effect of transposition is that each conductor has the same average inductance

Let line currents be,

$$I_A = I(\cos 0 + j\sin 0) = I(1 + j0)$$
$$I_B = I(\cos(-120) + j\sin(-120)) = I(-0.5 - j0.866)$$
$$I_C = I(\cos(120) + j\sin(120)) = I(-0.5 + j0.866)$$

Total flux linkages in A,

$$\psi_A = \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I_A - I_B \ln d_3 - I_C \ln d_2\right]$$

putting values of I_A, I_B & I_C

$$\psi_A = \frac{\mu_0}{2\pi}\left[\left(\frac{1}{4} - \ln r\right) I(1) - I(-0.5 - j0.866) \ln d_3 - I(-0.5 + j0.866) \ln d_2\right]$$

$$= \frac{\mu_0}{2\pi}\left[\frac{1}{4} I - I \ln r + 0.5 I_A + j\, I\, 0.866 \ln d_3 + 0.5 I \ln d_2 - j\, I\, 0.866 \ln d_2\right]$$

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$$= \frac{\mu_0}{2\pi}\left[\frac{1}{4}I - I\ln r + 0.5I(\ln d_3 + \ln d_2) + j0.866 I(\ln d_3 - \ln d_2)\right]$$

$$= \frac{\mu_0}{2\pi}\left[\frac{1}{4}I - I\ln r + I\ln\sqrt{d_3 d_2} + j0.866 I \ln\frac{d_3}{d_2}\right]$$

$$= \frac{\mu_0 I}{2\pi}\left[\frac{1}{4} + \ln\frac{\sqrt{d_3 d_2}}{r} + j0.866\ln\frac{d_3}{d_2}\right]$$

$$L_A = \frac{\mu_0}{2\pi}\left[\frac{1}{4} + \ln\frac{\sqrt{d_2 d_3}}{r} + j0.866\ln\frac{d_3}{d_2}\right]$$

Similarly, $L_B = \frac{\mu_0}{2\pi}\left[\frac{1}{4} + \ln\frac{\sqrt{d_3 d_1}}{r} + j0.866\ln\frac{d_1}{d_3}\right]$

$$L_C = \frac{\mu_0}{2\pi}\left[\frac{1}{4} + \ln\frac{\sqrt{d_1 d_2}}{r} + j0.866\ln\frac{d_2}{d_1}\right]$$

Inductance of each line conductor

$$= \frac{1}{3}(L_A + L_B + L_C)$$

$$= \frac{1}{3}\frac{\mu_0}{2\pi}\left[\frac{3}{4} + \ln\frac{d_1 d_2 d_3}{r}\right]$$

$$= \frac{\mu_0}{2\pi}\left[\frac{1}{4} + \ln\frac{\sqrt[3]{d_1 d_2 d_3}}{r}\right]$$

$$= \frac{4\pi\times10^{-7}}{2\pi}\left[\frac{1}{4} + \ln\frac{\sqrt[3]{d_1 d_2 d_3}}{r}\right]$$

$$= 2\times10^{-7}\left[\ln e^{1/4} + \ln\frac{\sqrt[3]{d_1 d_2 d_3}}{r}\right]$$

$$= 2\times10^{-7}\ln\frac{\sqrt[3]{d_1 d_2 d_3}}{r'}$$

Concept of Self-GMD & Mutual-GMD

↳ Used for simplifying the inductance calculations
  ↳ Specifically for multi-conductor arrangements.

(i) Self-GMD (D_s):

↳ Also known as GMR (Geometric mean radius)

· Inductance per conductor for wire $= 2\times10^{-7}\left(\frac{1}{4} + \ln\frac{d}{r}\right)$

↳ The term $2\times10^{-7}(1/4)$ is inductance due to flux within the solid cond.
↳ Desirable to eliminate this term by sig. self-GMD & GMR
↳ Replace the original by an equiv. hollow conductor with extremely thin walls,
↳ Current is confined to the surface and internal flux linkage is 0
↳ The radius of this equiv. cyl. should be sufficiently small
↳ To allow room for additional flux to compensate for internal flux.

$$\Rightarrow 2\times10^{-7}\left(\frac{1}{4} + \ln\frac{d}{r}\right)$$

$$= 2\times10^{-7}\left(\ln e^{1/4} + \ln\frac{d}{r}\right)$$

$$= 2\times10^{-7}\ln\frac{d}{e^{1/4}r}$$

$$= 2\times10^{-7}\ln\frac{d}{0.7788r} = 2\times10^{-7}\ln\frac{d}{r'}$$

where r' is the GMR and is $0.7788r$

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↳ Independent of the spacing b/w conductors
↳ Dependent on the size & shape of the conductors.

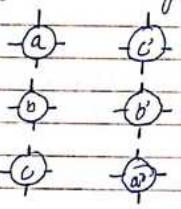
(ii) Mutual GMD:

↳ Geometrical mean of the distances from one conductor to the other
↳ Simply represents the ~~actual spacing b/w conductors~~ Equivalent Geometrical spacing

4(a) Mutual GMD b/w two conductors
↳ Distance b/w their centres
$D_m = d$

4(b) Mutual GMD for single ckt 3ϕ line
↳ Equivalent equilateral spacing
$D_m = (d_1 d_2 d_3)^{1/3}$

4(c) Mutual GMD for double ckt 3ϕ line



Self GMD of conductor = 0.7788r
Self GMD of combination aa' is
$D_{s1} = (D_{aa} \times D_{aa'} \times D_{a'a} \times D_{a'a'})^{1/4}$
Self GMD of combination bb' is
$D_{s2} = (D_{bb} \times D_{bb'} \times D_{b'b} \times D_{b'b'})^{1/4}$
Self GMD of combination cc' is
$D_{s3} = (D_{cc} \times D_{cc'} \times D_{c'c} \times D_{c'c'})^{1/4}$

Equivalent Self-GMD of one phase:
$D_s = (D_{s1} \times D_{s2} \times D_{s3})^{1/3}$

Mutual GMD b/w AϕB:
$D_{AB} = (D_{ab} \times D_{ab'} \times D_{a'b} \times D_{a'b'})^{1/4}$

Mutual GMD b/w BϕC:
$D_{BC} = (D_{bc} \times D_{bc'} \times D_{b'c} \times D_{b'c'})^{1/4}$

Mutual GMD b/w AϕC:
$D_{AC} = (D_{ac} \times D_{ac'} \times D_{a'c} \times D_{a'c'})^{1/4}$

Equivalent Mutual GMD, $D_m = (D_{AB} \times D_{BC} \times D_{AC})^{1/3}$

Inductance Formulas in terms of GMD

↳ (i) Single phase line:
Inductance/conductor/m $= 2 \times 10^{-7} \ln \frac{D_m}{D_s}$

$D_s = 0.7788r$
D_m = Spacing b/w the conductors $= d$

↳ (ii) Single ckt 3ϕ line:
Inductance/~~conductor~~ phase/m $= 2 \times 10^{-7} \ln \frac{D_m}{D_s}$

$D_s = 0.7788r$
$D_m = (d_1 d_2 d_3)^{1/3}$

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↳ (iii) Double ckt 3φ line:

Inductance/phase/m $= 2 \times 10^{-7} \ln \frac{D_m}{D_s}$

$D_s = (D_{s_1} D_{s_2} D_{s_3})^{1/3}$

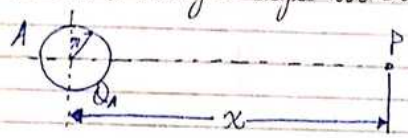
$D_m = (D_{AB} \times D_{BC} \times D_{AC})^{1/3}$

Electric Potential

↳ Work done in bringing a unit positive charge from infinity to a point

↳ Electric potential at a point due to a charge.

↳ Potential of a charged single conductor:



Electric Intensity E at a distance x,

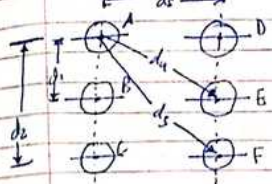
$$E = \frac{Q_A}{2\pi x \varepsilon_0} \text{ V/m}$$

where, Q_A = charge per metre length

ε_0 = Permittivity of free space

$$V_A = \int_x^\infty \frac{Q_A}{2\pi x \varepsilon_0} dx = \boxed{\frac{Q_A}{2\pi\varepsilon_0} \int_x^\infty \frac{dx}{x}}$$

↳ Potential of a conductor in a group of charged conductors:



↳ A, B, C with charges Q_A, Q_B, Q_C

Potential at A due to own charge:

$$= \int_r^\infty \frac{Q_A}{2\pi x \varepsilon_0} dx \qquad -(i)$$

Potential at A due to Q_B:

$$= \int_{d_1}^\infty \frac{Q_B}{2\pi x \varepsilon_0} dx \qquad -(ii)$$

Potential at A due to Q_C

$$= \int_{d_2}^\infty \frac{Q_C}{2\pi x \varepsilon_0} dx \qquad -(iii)$$

...

$V_A = (i) + (ii) + (iii) + \dots$

$$= \int_r^\infty \frac{Q_A}{2\pi x \varepsilon_0} dx + \int_{d_1}^\infty \frac{Q_B}{2\pi x \varepsilon_0} dx + \int_{d_2}^\infty \frac{Q_C}{2\pi x \varepsilon_0} dx + \dots$$

$$= \frac{1}{2\pi\varepsilon_0} \left[Q_A (\ln\infty - \ln r) + Q_B (\ln\infty - \ln d_1) + Q_C (\ln\infty - \ln d_2) + \dots \right]$$

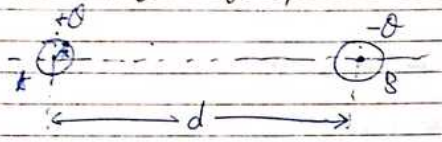
$$= \frac{1}{2\pi\varepsilon_0} \left[Q_A \ln\frac{1}{r} + Q_B \ln\frac{1}{d_1} + Q_C \ln\frac{1}{d_2} + \dots + \ln\infty (Q_A + Q_B + Q_C + \dots) \right]$$

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Assuming balanced condition, $Q_A + Q_B + Q_C \dots = 0$

$$V_A = \frac{1}{2\pi\varepsilon_0}\left[Q_A \ln\frac{1}{r} + Q_B \ln\frac{1}{d_1} + Q_C \ln\frac{1}{d_2} + \dots\right]$$

Capacitance of a Single phase Two-wire line



Total P.d. in conductor A,

$$V_A = \int_r^\infty \frac{Q\,dx}{2\pi x\varepsilon_0} + \int_d^\infty \frac{-Q}{2\pi x\varepsilon_0}\,dx$$

$$= \frac{Q}{2\pi\varepsilon_0}\left[\ln\frac{\infty}{r} - \ln\frac{\infty}{d}\right] = \frac{Q}{2\pi\varepsilon_0}\ln\frac{d}{r}\ V$$

Total P.d in conductor B,

$$V_B = \int_r^\infty \frac{-Q}{2\pi x\varepsilon_0}\,dx + \int_d^\infty \frac{Q}{2\pi x\varepsilon_0}\,dx$$

$$= \frac{-Q}{2\pi\varepsilon_0}\ln\frac{d}{r}\ V$$

Total P.d b/w conductor, $V_{AB} = V_A - V_B$

$$= 2V_A = \frac{Q}{\pi\varepsilon_0}\ln\frac{d}{r}\ V$$

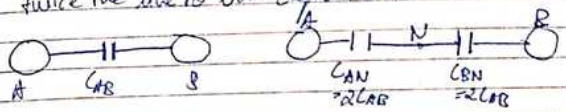
Capacitance, $C_{AB} = Q/V_{AB} = \dfrac{Q}{\frac{Q}{\pi\varepsilon_0}\ln\frac{d}{r}} = \boxed{\dfrac{\pi\varepsilon_0}{\ln\frac{d}{r}}\ F/m}$

↳ Capacitance to neutral:

↳ Capacitance b/w one of the conductors and a neutral point is known

↳ Since, potential at midpoint is zero

↳ Capacitance to ground or neutral for two wire line is twice the line to line capacitance



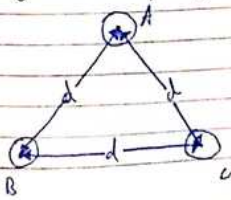
$$C_N = C_{AN} = C_{BN} = 2C_{AB}$$

$$\boxed{C_N = \frac{2\pi\varepsilon_0}{\ln\frac{d}{r}}\ F/m}$$

Capacitance of a 3ϕ overhead line

↳ Capacitance of each conductor is considered instead of capacitance from conductor to conductor

4(i) Symmetrical Spacing:



↳ Conductors A, B, C

↳ Charges Q_A, Q_B, Q_C

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Overall potential difference at h_1

$$V_A = \int_h^d \frac{Q_A}{2\pi\varepsilon_0 x}dx + \int_d^D \frac{Q_B}{2\pi\varepsilon_0 x}dx + \int_D^\infty \frac{Q_C}{2\pi\varepsilon_0 x}dx$$

$$= \frac{1}{2\pi\varepsilon_0}\left[Q_A \ln\frac{1}{h} + Q_B \ln\frac{1}{d} + Q_C \ln\frac{1}{D}\right]$$

$$= \frac{1}{2\pi\varepsilon_0}\left[Q_A \ln\frac{1}{h} + (Q_B + Q_C)\ln\frac{1}{d}\right]$$

Assuming balanced condition, $Q_A + Q_B + Q_C = 0$

$$= \frac{1}{2\pi\varepsilon_0}\left[Q_A \ln\frac{1}{h} - Q_A \ln\frac{1}{d}\right]$$

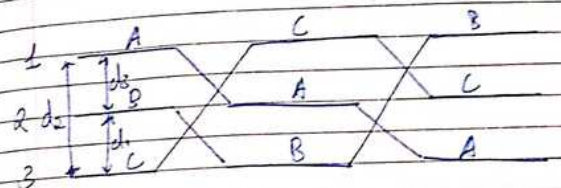
$$V_A = \frac{Q_A}{2\pi\varepsilon_0}\ln\frac{d}{h}\ V$$

Capacitance, $C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi\varepsilon_0}\ln\frac{d}{h}} = \frac{2\pi\varepsilon_0}{\ln\frac{d}{h}}$ F/m

$$\boxed{C_A = \frac{2\pi\varepsilon_0}{\ln\frac{d}{h}}\ \text{F/m}}$$

∴ Similar for B & C

(ii) Unsymmetrical Spacing :



∴ For phase A,

Potential for 1, $V_1 = \frac{1}{2\pi\varepsilon_0}\left(Q_A \ln\frac{1}{r} + Q_B \ln\frac{1}{d_2} + Q_C \ln\frac{1}{d_3}\right)$

Potential for 2, $V_2 = \frac{1}{2\pi\varepsilon_0}\left(Q_A \ln\frac{1}{r} + Q_B \ln\frac{1}{d_1} + Q_C \ln\frac{1}{d_2}\right)$

Potential for 3, $V_3 = \frac{1}{2\pi\varepsilon_0}\left(Q_A \ln\frac{1}{r} + Q_B \ln\frac{1}{d_2} + Q_C \ln\frac{1}{d_1}\right)$

Average voltage of h_1

$$V_A = \frac{1}{3}(V_1 + V_2 + V_3)$$

$$= \frac{1}{3\times 2\pi\varepsilon_0}\left(Q_A \ln\frac{1}{r^3} + (Q_B + Q_C)\ln\frac{1}{d_1 d_2 d_3}\right)$$

$$= \frac{1}{3\times 2\pi\varepsilon_0}\left(Q_A \ln\frac{1}{r^3} - Q_A \ln\frac{1}{d_1 d_2 d_3}\right)$$

[Since, $Q_A + Q_B + Q_C = 0$, $Q_B + Q_C = -Q_A$]

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$$V_A = \frac{Q_A}{6\pi\varepsilon_0}\left[\ln\frac{d_1 d_2 d_3}{R^3}\right]$$

$$V_A = \frac{Q_A}{2\pi\varepsilon_0}\ln\frac{\sqrt[3]{d_1 d_2 d_3}}{R^3}$$

Capacitance, $C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi\varepsilon_0}\ln\frac{\sqrt[3]{d_1 d_2 d_3}}{R^3}}$

$$C_A = \frac{2\pi\varepsilon_0}{\ln\frac{\sqrt[3]{d_1 d_2 d_3}}{R^3}} \text{ F/m}$$

Classification of Overhead Transmission line

↳ Transmission line has 3 constants (i) R (ii) L (iii) C distributed uniformly along the whole length of the line
↳ Resistance and inductance forms the series impedance
↳ Capacitance (conductor to conductor in 1ϕ2 conductor to neutral in 3ϕ line) forms a shunt path throughout the length

↳ Depending upon how the capacitance is taken into account, often

↳ (i) Short transmission line:

↳ Length upto about 50 km
↳ Line voltage is comparatively low (<20kV)
↳ Due to smaller length & voltage capacitance is negligible
↳ Only R & L are taken into account

↳ (ii) Medium transmission line:

↳ Length about 50 - 150 km
↳ Line voltage moderately high (>20kV <100kV)
↳ Capacitance effects are taken into account
↳ Divided and lumped in the form of condensers shunted across the line

↳ (iii) Long transmission line:

↳ Length >150 km
↳ Line voltage >100kV
↳ Line constants are uniformly distributed across the whole length

Some important terms

↳ (i) Voltage Regulation:

↳ When transmission line is carrying current there is a voltage drop due to resistance and inductance of the line
↳ Recieving end voltage (V_R) < Sending end voltage (V_S)
↳ This voltage drop ($V_S - V_R$) is expressed as a % age of V_R
↳ This is known as voltage regulation
↳ The difference in voltage at the recieving end of transmission line between conditions of no load and full load is called voltage regulation

$$\%\text{age Voltage regulation} = \frac{V_S - V_R}{V_R} \times 100$$

↳ Voltage regulation should be low.

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↳(ii) Transmission Efficiency:

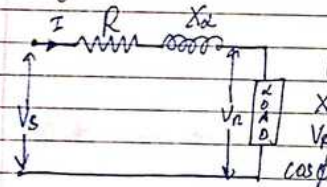
↳ Power received at receiving end is generally less than the power at sending end due to losses

% transmission efficiency, η_T = Receiving end power / Sending end power ×100

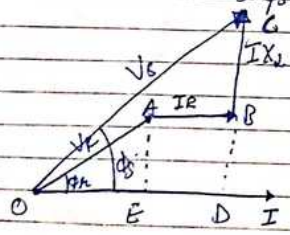
$$= \frac{V_R I_R \cos\phi_R}{V_S I_S \cos\phi_S} \times 100$$

Performance of Single Phase Short transmission lines

↳ Only resistance and inductance are taken into account



I = load current
R = loop resistance
X_L = loop resistance
V_R = Receiving end voltage
$\cos\phi_R$ = receiving end power factor (lagging)
V_S = sending end voltage
$\cos\phi_S$ = sending end power factor



From ODC,

$$OC^2 = OD^2 + DC^2$$
$$V_S^2 = (OE+ED)^2 + (DB+BC)^2$$
$$V_S^2 = (V_R\cos\phi_R + IR)^2 + (V_R\sin\phi_R + IX_L)^2$$
$$\boxed{V_S = \sqrt{(V_R\cos\phi_R + IR)^2 + (V_R\sin\phi_R + IX_L)^2}}$$

↳(i) % Voltage Regulation = $\frac{V_S - V_R}{V_R} \times 100$

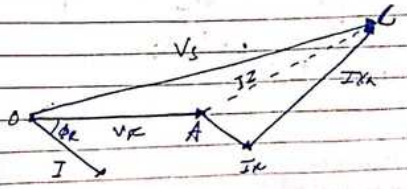
↳(ii) Sending end pf, $\cos\phi_S = \frac{OD}{OC} = \frac{V_R\cos\phi_R + IR}{V_S}$

↳(iii) Power delivered = $V_R I_R \cos\phi_R$
Line loss = I^2R
Power sent out = $V_R I_R \cos\phi_R + I^2R$

% age transmission efficiency = Power delivered / Power sent out ×100

$$= \frac{V_R I_R \cos\phi_R}{V_R I_R \cos\phi_R + I^2R} \times 100$$

↳ Solution in complex notation:
↳ Taking $\vec{V_R}$ as reference



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$\vec{V_R} = V_R + 0j$

$\vec{I} = I\angle -\phi_R = I(\cos\phi_R - j\sin\phi_R)$

$\vec{Z} = R + jX_L$

$\vec{V_S} = \vec{V_R} + \vec{I}\vec{Z}$

$= (V_R + j0) + I(\cos\phi_R - j\sin\phi_R)(R + jX_L)$

$= (V_R + IR\cos\phi_R + IX_L\sin\phi_R) + j(IX_L\cos\phi_R - IR\sin\phi_R)$

$V_S = \sqrt{(V_R + IR\cos\phi_R + IX_L\sin\phi_R)^2 + (IX_L\cos\phi_R - IR\sin\phi_R)^2}$

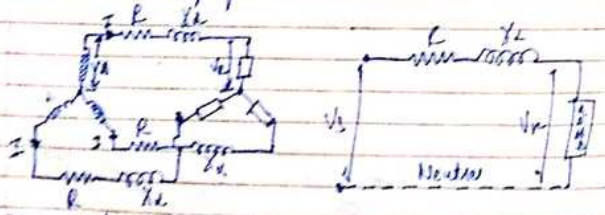
Small and negligible

$\Rightarrow \boxed{V_S = V_R + IR\cos\phi_R + IX_L\sin\phi_R}$

↳ Approximate formula and applicable for lagging pf
↳ Leading pf gives appreciable error

Three phase Short Transmission line

↳ Regarded as combination of 3 single phase units
↳ Voltage are phase voltage & R & X_L are the resistance and inductive reactance per phase



Y-connected generator supplying a Y-connected load

One phase

Effect of load pf on Regulation and Efficiency

↳ Regulation and efficiency depend upon the power factor of the load

↳(i) Effect on regulation:

$\%age\ regulation = \frac{IR\cos\phi_R + IX_L\sin\phi_R}{V_R} \times 100$ [For lagging PF]

$\%age\ regulation = \frac{IR\cos\phi_R - IX_L\sin\phi_R}{V_R} \times 100$ [For leading PF]

↳ When load pf is lagging or unity or such that $IR\cos\phi_R > IX_L\sin\phi_R$
↳ Voltage regulation is positive i.e. $V_S < V_R$
↳ Voltage regulation increases with decrease in pf of lagging loads

↳ When load is leading such that $IX_L\sin\phi_R > IR\cos\phi_R$
↳ Voltage regulation is negative i.e. $V_S < V_R$
↳ Voltage regulation decreases with decrease in pf of leading loads

↳(ii) Effect on transmission efficiency:

$P = V_R I\cos\phi_R$ [For 1ϕ]

$\Rightarrow I = \frac{P}{V_R\cos\phi_R}$

$P = \sqrt{3}V_R I\cos\phi_R$ [For 3ϕ]

$I = \frac{P}{\sqrt{3}V_R\cos\phi_R}$

↳ $I \propto \frac{1}{\cos\phi_R}$ for given amount of P & V_R
↳ Decrease in load pf ⇒ increase in load current ⇒ increase in line losses

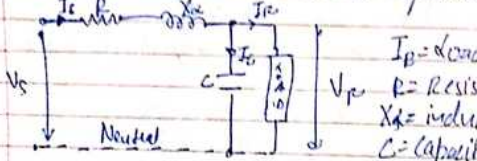
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Medium Transmission Lines

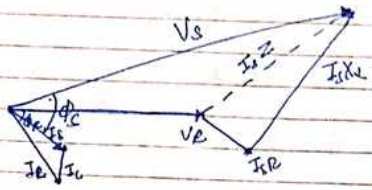
↳ Most commonly used methods for the solution for medium TL:
- (a) End Condenser Method
- (b) Nominal T method
- (c) Nominal π method

End Condenser Method

↳ Capacitance of the line is condensed & concentrated at the receiving or load end
↳ This method overestimates the capacitance



I_R = load current per phase
R = Resistance per phase
X_L = inductive reactance per phase
C = Capacitance per phase
$\cos\phi_R$ = receiving end power factor (lagging)



$\vec{V}_R = V_R + j0$
$\vec{I}_R = I_R(\cos\phi_R - j\sin\phi_R)$
$\vec{I}_c = j\vec{V}_R\omega C = j2\pi fC\vec{V}_R$
$\vec{I}_s = \vec{I}_R + \vec{I}_c$
$= I_R(\cos\phi_R - j\sin\phi_R) + j2\pi fC\vec{V}_R$

Voltage drop/phase $= \vec{I}_s\vec{Z} = \vec{I}_s(R + jX_L)$

$\vec{V}_s = \vec{V}_R + \vec{I}_s\vec{Z} = \vec{V}_R + \vec{I}_s(R + jX_L)$

% voltage regulation $= \frac{V_s - V_R}{V_R} \times 100$

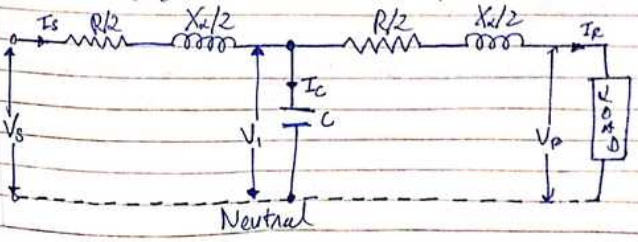
% transmission efficiency $= \frac{\text{Power delivered/phase}}{\text{Power del. per phase} + \text{Losses per phase}} \times 100$

$= \frac{V_R I_R \cos\phi_R}{V_R I_R \cos\phi_R + I_s^2 R} \times 100$

↳ Limitations: (i) Considerable error (10%) because
↳ distributed capacitance has been assumed to be lumped or concentrated
(ii) Overestimates the effects of line capacitance

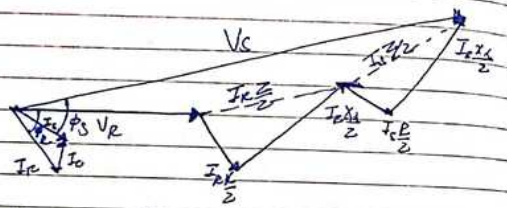
Nominal T Method

↳ The whole line capacitance is assumed to be concentrated at the middle point of the line & half the line resistance and capacitance are lumped on either side
↳ Full charging current flows over half the line



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I_R = Load Current per phase
X_L = Inductive reactance per phase
$\cos\phi_R$ = Recieving end power factor (lagging)
V_1 = Voltage across capacitor C
R = Resistance per phase
C = Capacitance per phase
V_S = Sending end voltage/phase



Recieving end voltage, $\vec{V}_R = V_R + j0$

load current, $\vec{I}_R = I_R(\cos\phi_R - j\sin\phi_R)$

Voltage across C, $\vec{V}_1 = \vec{V}_R + \vec{I}_R\vec{Z}/2$

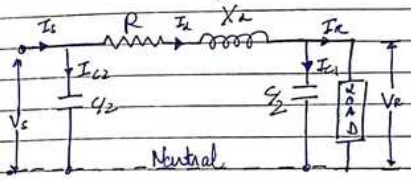
$= \vec{V}_R + I_R(\cos\phi_R - j\sin\phi_R)\left(\frac{R}{2} + j\frac{X_L}{2}\right)$

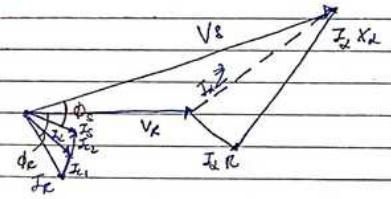
Capacitive current, $\vec{I}_C = \vec{I}_R + \vec{I}_C$

Sending end voltage, $\vec{V}_S = \vec{V}_1 + \frac{\vec{I}_S\vec{Z}}{2} = \vec{V}_1 + \vec{I}_S\left(\frac{R}{2} + j\frac{X_L}{2}\right)$

Nominal π Method

↳ Capacitance divided into two halves, one half lumped at the sending end and other half lumped at the recieving end
↳ Capacitance at sending end has no effect on line loss
↳ However its charging current must be added to line current to obtain total sending current





I_R = load current per phase
R = Resistance per phase
X_L = Inductive reactance per phase
C = Capacitance per phase
$\cos\phi_R$ = Recieving end power factor (lagging)
V_S = Sending end voltage per phase

$\vec{V}_R = V_R + j0$
load current, $\vec{I}_R = I_R(\cos\phi_R - j\sin\phi_R)$
charging current at load end is, $\vec{I}_C = j\omega(C/2)\vec{V}_R = j\pi fC\vec{V}_R$

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line current, $\vec{I}_R = \vec{I}_{R'} + \vec{I}_{C_1}$

Sending end voltage, $\vec{V}_S = \vec{V}_{R'} + \vec{I}_R \vec{Z} = \vec{V}_{R'} + \vec{I}_R (R + jX_L)$

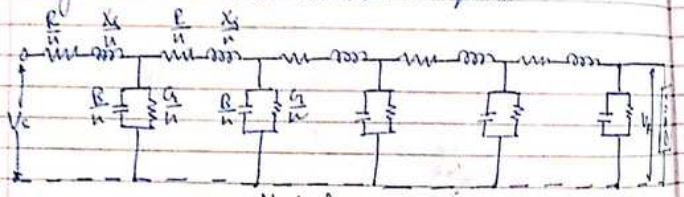
Charging current at the sending end is,

$$\vec{I}_{C_2} = j\omega(C/2)\vec{V}_S = j\omega C \vec{V}_S$$

Sending current, $\vec{I}_S = \vec{I}_R + \vec{I}_{C_2}$

Long Transmission Lines

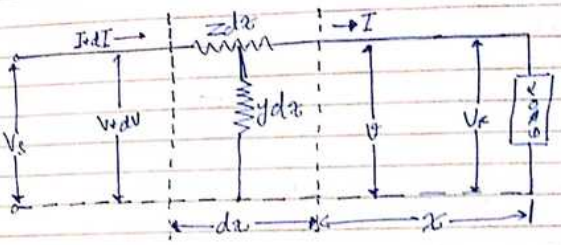
↳ Constants cannot be assumed as lumped
↳ Rigorous mathematical treatment is required



↳ Points to be noted: (i) The line constants are uniformly distributed over the entire length of the line
(ii) Resistance and inductive reactance are the series elements
(iii) Leakage susceptance (B) & leakage conductance (G) are shunt elements
↳ Leakage susceptance is due to the fact that capacitance exists b/w line & neutral
↳ Leakage conductance takes into account the power loss is occuring through leakage over the insulators or due to corona.

↳ Admittance = $\sqrt{G^2+B^2}$
(iv) The leakage current through shunt admittance is maximum at the sending end & decreases continuously as the receiving end is approached at which point it is zero.

Analysis of long Transmission line (Rigorous Method)



↳ Considering a small element in the line of length dx at a distance x from receiving end

z = series impedance of the line per unit length
y = shunt admittance of the line per unit length
V = Voltage at the end of element towards receiving end
$V+dV$ = Voltage at the end of element towards sending end
$I+dI$ = current entering the element dx
I = current leaving the element dx
↳ Then, for the small element dx,

$z\,dx$ = Series impedance
$y\,dx$ = shunt admittance
Obviously, $dV = Iz\,dx$

$$\frac{dV}{dx} = Iz \qquad -(i)$$

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dI = Current through shunt admittance of element
$= Vy\,dx$

$$\frac{dI}{dx} = Vy \qquad \text{(ii)}$$

diff eq(i) wrt x,

$$\frac{d^2V}{dx^2} = z\frac{dI}{dx} = z(Vy)$$

$$\frac{d^2V}{dx^2} = yzV \qquad \text{(iii)}$$

Solution of this differential Equation:

$$V = K_1\cosh(x\sqrt{yz}) + K_2\sinh(x\sqrt{yz}) \qquad \text{(iv)}$$

diff. wrt x,

$$\frac{dV}{dx} = K_1\sqrt{yz}\sinh(x\sqrt{yz}) + K_2\sqrt{yz}\cosh(x\sqrt{yz})$$

But, $\frac{dV}{dx} = I_x$

$$I_x = K_1\sqrt{yz}\sinh(x\sqrt{yz}) + K_2\sqrt{yz}\cosh(x\sqrt{yz})$$

$$I = \sqrt{\frac{y}{z}}\left[K_1\sinh(x\sqrt{yz}) + K_2\cosh(x\sqrt{yz})\right] \qquad \text{(v)}$$

Eq(iv) & (v) give V & I in terms of K_1 & K_2
↳ finding K_1 & K_2

At $x=0$, $V=V_R$, $I=I_R$ in eq(iv)

$$V_R = K_1\cosh 0 + K_2\sinh 0$$
$$= K_1$$

At $x=0$, $V=V_R$, $I=I_R$ in eq(v)

$$I_R = \sqrt{\frac{y}{z}}\left[K_1(0) + K_2\right]$$

$$I_R = \sqrt{\frac{y}{z}}K_2$$

$$K_2 = \sqrt{\frac{z}{y}}I_R$$

Substituting

$$V = V_R\cosh(x\sqrt{yz}) + \sqrt{\frac{z}{y}}I_R\sinh(x\sqrt{yz})$$

$$I = \sqrt{\frac{y}{z}}V_R\sinh(x\sqrt{yz}) + I_R\cosh(x\sqrt{yz})$$

Sending end voltage V_S & sending end current I_S can be found by putting $x=l$

$$V_S = V_R\cosh(l\sqrt{yz}) + \sqrt{\frac{z}{y}}I_R\sinh(l\sqrt{yz})$$

$$I_S = \sqrt{\frac{y}{z}}V_R\sinh(l\sqrt{yz}) + I_R\cosh(l\sqrt{yz})$$

Now,

$$l\sqrt{yz} = \sqrt{l^2yz} = \sqrt{YZ}$$

& $\sqrt{\frac{y}{z}} = \sqrt{\frac{l^2y}{l^2z}} = \sqrt{\frac{Y}{Z}}$

Y = total shunt admittance of tx line
Z = total series impedance of tx line

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Therefore,

$$V_s = V_r \cosh(\sqrt{YZ}) + I_r \sqrt{\frac{Z}{Y}} \sinh(\sqrt{YZ})$$

$$I_s = V_r \sqrt{\frac{Y}{Z}} \sinh\sqrt{YZ} + I_r \cosh(\sqrt{YZ})$$

↳ It is helpful to expand hyperbolic sine & cosine

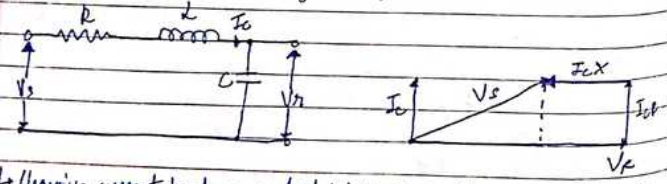
$$\cosh(\sqrt{YZ}) = \left(1 + \frac{ZY}{2} + \frac{Z^2Y^2}{2n} + \dots\right)$$

$$\sinh\sqrt{YZ} = \left(\sqrt{YZ} + \frac{(YZ)^{3/2}}{6} + \dots\right)$$

Ferranti Effect

↳ When a long line is operating in no load or light load
↳ the receiving end voltage is greater than sending end voltage
↳ This is known as ferranti effect

↳ Since, usually the capacitive reactance is quite large as compared to inductive reactance
↳ Under no load or light load conditions
↳ line current is of leading power factor



↳ Charging current produces a drop which is in phase opposition to the receiving end voltage
↳ Hence, sending end voltage becomes smaller than receiving end

Surge Impedance loading

↳ The line generates capacitive reactive VA in its shunt capacitance & absorbing reactive VA in its series inductance.
↳ The load at which the inductive & reactive VA are equal and opposite is called surge impedance load

↳ If the loading is less than SIL, ferranti effect occurs

Insulated Cables

↳ Electric cables consist of 3 essential parts
↳ (a) The conductor for transmitting electrical power
↳ (b) The insulation, an electric insulating medium, needed to insulate the conductor from direct contact with earth or other objects
↳ (c) External protection against mechanical damage, chemical & electro-chemical attack, fire etc.

Main requirements for the insulating materials used

↳ (i) High insulation resistance
↳ (ii) high dielectric strength
↳ (iii) Good mechanical properties i.e. tenacity and elasticity
↳ (iv) It should not be affected by chemicals around it
↳ (v) It should not be non-hygroscopic
↳ Dielectric strength of any material goes down with moisture content

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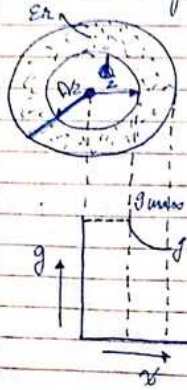
Classification of Cables

- Cables are classified in two ways
  - (i) Type of insulating material used
  - (ii) Voltage for which they are manufactured
- (ii) is generally preferred.

- On the basis of voltage they are:
  - (i) Low tension cables (L.T.) – upto 1000 V
  - (ii) High tension cables (H.T.) – upto 11000 V
  - (iii) Super tension cables (S.T.) – from 22kV to 33 kV
  - (iv) Extra High Tension Cables (E.H.T.) – from 33kV to 66kV
  - (v) Extra Super Voltage Cables – beyond 132 kV

Dielectric Stress in a Single-Core Cable

- Under operating conditions the insulation is subjected to electrostatic forces
  - This is known as dielectric stress
  - Potential gradient at that point



Core diameter, d
Internal sheath diameter D

Electrical intensity at point x m from the inner

$$E_x = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x} \text{ V/m}$$

Potential gradient g = Electrical intensity

$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x} \text{ V/m}$$

Potential diff. V b/w conductor and sheath

$$V = \frac{Q}{2\pi\varepsilon_0\varepsilon_r} \ln\frac{D}{d} \text{ Volts}$$

$$Q = \frac{2\pi\varepsilon_0\varepsilon_r V}{\ln\frac{D}{d}}$$

Substituting for g

$$g = \frac{2\pi\varepsilon_0\varepsilon_r V}{\ln\frac{D}{d}} \times \frac{1}{2\pi\varepsilon_0\varepsilon_r x}$$

$$g = \frac{V}{x \ln\frac{D}{d}} \text{ V/m}$$

Maximum potential gradient, x = d/2

$$g_{max} = \frac{2V}{d \ln\frac{D}{d}} \text{ V/m}$$

Minimum potential gradient, x = D/2

$$g_{min} = \frac{2V}{D \ln\frac{D}{d}} \text{ V/m}$$

$$\frac{g_{max}}{g_{min}} = \frac{\frac{2V}{d \ln\frac{D}{d}}}{\frac{2V}{D \ln\frac{D}{d}}} = \frac{D}{d}$$

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Grading of Cables

↳ The process of achieving uniform electrostatic stress in the dielectric of cables.
↳ Grading of cables

↳ Electrostatic stress in a single core cable has max value (g_{max}) at the conductor surface and goes on decreasing towards the sheath
↳ For safe working the strength of dielectric must be more than g_{max}
↳ High dielectric strength is only useful near the conductor
↳ as we move away from the conductor, electrostatic stress decreases
↳ Dielectric will be unnecessarily overstrung
↳ Undesirable for two reasons
↳ (i) Insulation of greater thickness is required
↳ increase in cable size
(ii) It may lead to the breakdown of insulation
↳ It is necessary to have uniform stress distribution
↳ Value is increased in the outer layers of the dielectric
↳ Two main methods:
↳ (a) Capacitance grading
↳ (b) Intercream grading

↳ Capacitance Grading

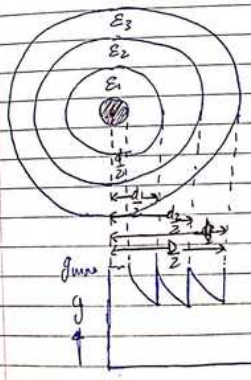
↳ Achieving uniformity in the dielectric stress by using layers of different dielectrics
↳ Capacitance grading
↳ Homogeneous dielectric is replaced by a composite dielectric
↳ consists of various layers of different dielectrics
↳ Relative permittivity ε_r of each layer is inversely proportional to its distance from the centre
↳ Potential gradient is constant and independent of distance from the centre

↳ As $\varepsilon_r \propto \frac{1}{x} \Rightarrow \varepsilon_r = \frac{K}{x}$ [K is a constant]

Potential gradient at distance x from the centre.

$$= \frac{Q}{2\pi\varepsilon_0\varepsilon_r x} = \frac{Q}{2\pi\varepsilon_0 (K/x) x} = \frac{Q}{2\pi\varepsilon_0 K} = \text{constant}$$

↳ Ideal grading requires infinite no. of dielectrics
↳ In practice two or three are used in decreasing order of permittivity
↳ Dielectric of highest permittivity used near the core



$$g_{1max} = \frac{Q}{2\pi\varepsilon_0\varepsilon_1 d_1} \; ; \; g_{2max} = \frac{Q}{2\pi\varepsilon_0\varepsilon_2 d_1} \; ; \; g_{3max} = \frac{Q}{2\pi\varepsilon_0\varepsilon_3 d_2}$$

$\therefore g_{1max} = g_{2max} = g_{3max} = g_{max}$ (say)

$$\frac{1}{\varepsilon_1 d} = \frac{1}{\varepsilon_2 d_1} = \frac{1}{\varepsilon_3 d_2}$$

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$\Rightarrow \varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$

Potential diff. across the inner layer is

$$V_1 = \int_{d/2}^{d_1/2} g\,dx = \int_{d/2}^{d_1/2} \frac{Q}{2\pi\varepsilon_0\varepsilon_1 x}\,dx$$

$$= \frac{Q}{2\pi\varepsilon_0\varepsilon_1} \ln\frac{d_1}{d} \quad \left[\because \frac{g_{max}\,d}{2} = \frac{Q}{2\pi\varepsilon_0\varepsilon_1}\right]$$

$$= \frac{g_{max}\,d}{2} \ln\frac{d_1}{d}$$

Similarly,

$$V_2 = \frac{g_{max}}{2} d_1 \ln\frac{d_2}{d_1}$$

$$V_3 = \frac{g_{max}}{2} d_2 \ln\frac{D}{d_2}$$

Total P.d b/w core & earthed sheath,

$$V = V_1 + V_2 + V_3$$

$$= \frac{g_{max}}{2}\left[d \ln\frac{d_1}{d} + d_1 \ln\frac{d_2}{d_1} + D \ln\frac{D}{d_2}\right]$$

↳ If homogeneous dielectric

$$V' = \frac{g_{max}}{2} d \ln\frac{D}{d}$$

Hence $\Rightarrow V > V'$

↳ Cross cable can be operated at a greater potential than non graded

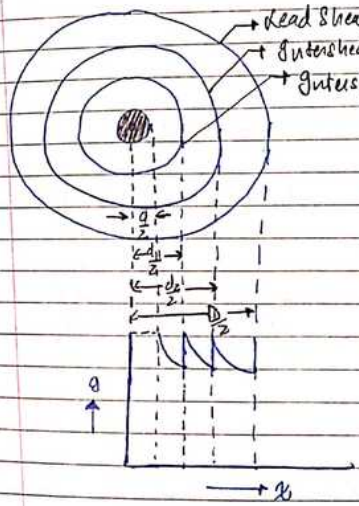
↳ The principle disadvantage is that, a few high graded dielectrics of reasonable cost have permittivities that vary over the required range.

2. Intersheath Grading

↳ A homogeneous dielectric is used but is divided into layers by placing metallic intersheath b/w core & lead sheath.

↳ Metallic sheaths are held at suitable potential

↳ in between core potential and earth potential.



Maximum stress b/w core & intersheath

$$g_{1max} = \frac{2V_1}{d \ln\frac{d_1}{d}}$$

Similarly,

$$g_{2max} = \frac{2V_2}{d_1 \ln\frac{d_2}{d_1}}$$

$$g_{3max} = \frac{2V_3}{d_2 \ln\frac{D}{d_2}}$$

And, $g_{1max} = g_{2max} = g_{3max} = g_{max}$ (say)

$$\frac{2V_1}{d \ln\frac{d_1}{d}} = \frac{2V_2}{d_1 \ln\frac{d_2}{d_1}} = \frac{2V_3}{d_2 \ln\frac{D}{d_2}}$$

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↳ As the cable behaves like 5 capacitors in series
↳ All potentials are in phase
↳ Voltage of conductor & lead sheath

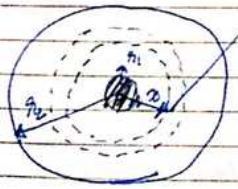
$V = V_1 + V_2 + V_3$

↳ Principle disadvantages
↳ (i) Complications in fixing the sheath potentials
↳ (ii) Intersheaths are likely to be damaged during transportation & use
↳ might result in local concentrations of potential gradient
↳ (iii) Considerable losses in the intersheaths due to charging current.

↳ Intersheath grading is rarely used

Insulation Resistance of a Single-Core Cable

↳ Cable is provided with insulating material of suitable thickness to avoid leakage current
↳ The path for leakage current is radial through the insulation
↳ The opposition provided by the insulation to the leakage current is known as insulation resistance of the cable
↳ For satisfactory operation, insulation resistance is very high



Insulation resistance of considered layer

$= \frac{\rho\, dx}{2\pi x l}$

Insulation resistance of the whole cable,

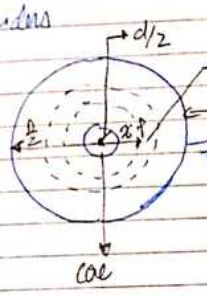
$R = \int_{r_1}^{r_2} \frac{\rho\, dx}{2\pi x l} = \frac{\rho}{2\pi l} \int_{r_1}^{r_2} \frac{1}{x}\, dx$

$R = \frac{\rho}{2\pi l} \ln \frac{r_2}{r_1}$

↳ $R \propto \frac{1}{l}$

Capacitance of a Single-Core Cable

↳ Single-core cable can be considered to be equivalent to two long coaxial cylinders



Surface area of cylinder with radius x m & axial length 1m
$= 2\pi x \times 1\ m^2$

Electric flux density at any point distance x from the centre,

$D_x = \frac{Q}{2\pi x}\ C/m^2$

Electric field intensity at distance x,

$E_x = \frac{D_x}{\varepsilon} = \frac{Q}{2\pi\varepsilon_0 x} = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x}\ V/m$

Work done in moving a charge from x through a distance dx.
$E_x dx$
∴ Work done in moving a charge from conductor to sheath

$V = \int_{d/2}^{D/2} E_x\, dx = \int_{d/2}^{D/2} \frac{Q}{2\pi\varepsilon_0\varepsilon_r} \frac{1}{x}\, dx = \frac{Q}{2\pi\varepsilon_0\varepsilon_r} \ln \frac{D}{d}$

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Capacitance of the cable is,

$$C = \frac{Q}{V} = \frac{Q}{\frac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln\frac{D}{d}} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln D/d} \text{ F/m}$$

$$= \frac{2\pi \times 8.854\times10^{-12}\times\varepsilon_r}{2.303\log D/d} \text{ F/m}$$

$$= \frac{\varepsilon_r}{41.4\log D/d}\times10^{-9} \text{ F/m}$$

If cable length is l m,

$$C = \frac{\varepsilon_r l}{41.4\log\frac{D}{d}}\times10^{-9} \text{ F}$$

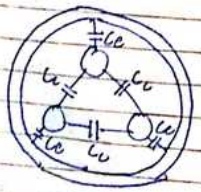
Capacitance of a 3-core Cable

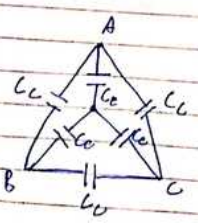
↳ Since potential difference exists b/w pairs of conductors & b/w each conductor and the sheath

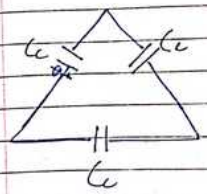
↳ Electrostatic fields are set up in the cable

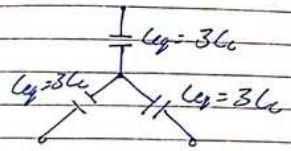
↳ these give rise to core-core capacitance (C_c) & conductor-sheath capacitance (C_e)

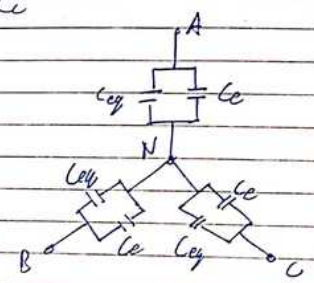
↳ C_c are delta connected & C_e are star connected

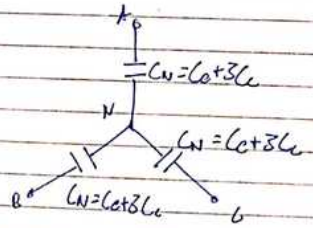












$C_N = C_e + C_{eq}$

$= C_e + 3C_c$

↳ If V_{ph} is the phase voltage, then charging current I_c.

$$I_c = \frac{V_{ph}}{\text{Capacitive reactance per phase}} = 2\pi f V_{ph} C_N$$

$$I_c = 2\pi f V_{ph}(C_e + 3C_c)$$

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→ Measurement of C_e and C_c

↳ (i) In the first measurement, 2 cores are bunched together and capacitance is measured b/w the bunched cores and the sheath

↳ Bunching eliminates all the three C_c capacitances

↳ ∴ If C_1 is the measured capacitance

$$C_1 = 3C_e$$

$$C_e = \frac{C_1}{3}$$

↳ (ii) In the second measurement, 2 cores are bunched with the sheath

↳ Capacitance is measured b/w them and the 3rd core

↳ This yields $2C_c + C_e$

↳ ∴ If C_2 is the measured capacitance

$$C_2 = 2C_c + C_e$$

↳ If C_N is required

↳ another test in which the capacitance b/w two cores or lines is measured with the third core free or connected to the sheath

↳ This eliminates one of the C_e

↳ ∴ If C_3 is the measured capacitance,

$$C_3 = C_c + \frac{C_e}{2} + \frac{C_e}{2}$$

$$= \frac{1}{2}(C_e + 2C_c)$$

$$= \frac{1}{2}C_N$$

Heating of Cables

↳ Sources of heat generation in cables are,

↳ (i) Core loss

↳ (ii) Dielectric loss

↳ (iii) Sheath losses

↳ (i) Core loss : ↳ To find core loss, resistance is calculated

↳ (i) by measuring the resistance at ambient temperature (20°C)

Assuming an operating temperature of 65°C

$$R_{65} = R_{20}(1 + \alpha t)$$

$$t = 65 - 20 = 45^\circ C$$

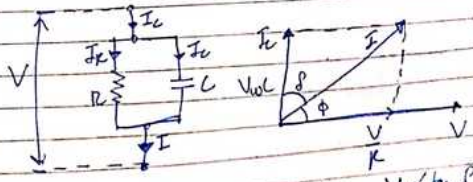
↳ (ii) Since the effective area of section of the cable is smaller than the actual physical section, effective resistance is larger

↳ A factor of 1.02 is multiplied to get the resistance

↳ Dielectric loss : ↳ The cable is a part of a capacitor

↳ Loss in the dielectric is due to the loss in equivalent leakage resistance,

$$P = \frac{V^2}{R}$$



$$\frac{V/R}{V_{NL}} = \tan\delta \Rightarrow \frac{V}{R} = V_{NL}\tan\delta$$

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$P = V^2 \omega C \tan\delta$

$\delta \rightarrow$ dielectric loss angle

Faults in a Power System

↳ Fault occurs when two or more conductors that normally operate with a potential difference come in contact with each other

↳ Hidden failure ~~failure~~ failure of equipment

↳ Accidental damage

↳ Short ckt due to insulation failure from lightening surges

↳ Classified into two main categories

↳ (i) Symmetrical Faults

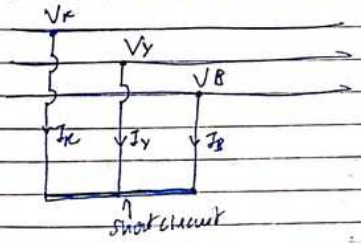
↳ (ii) Unsymmetrical Faults

Symmetrical Faults on 3phase system

↳ The fault which gives rise to symmetrical fault currents

↳ equal fault currents in the lines with 120° displacement

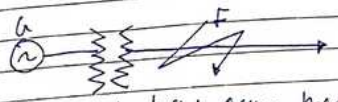
↳ Occurs when all the 3 conductors are brought together



↳ Rarely occurs, most are symmetrical

↳ Most severe and imposes more heavy duty on the circuit breaker

Limitation of fault current



↳ When a short ckt occurs, the short ckt current is limited by the impedance of the system upto the point of fault.

↳ If a fault occurs at point F, the short ckt current will be limited by

↳ impedance of generator, transformer and the impedance of the line from the generator upto the fault point

↳ Knowledge of impedances of various equipment + circuits in the line is important for determining short-circuit current

Percentage Reactance

↳ Reactance of generators, transformers, reactors etc is usually expressed in percentage reactance

↳ for rapid calculations.

↳ It is the percentage of the total phase-voltage dropped in the circuit when full-load current is flowing.

$$\% X = \frac{IX}{V} \times 100$$

I = full load current

V = phase voltage

X = reactance in ohms per phase

Alternatively,

$$\% X = \frac{(kVA)X}{10(kV)^2}$$ [X is the reactance in Ω]

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↳ If X is the only reactance in the circuit then

$$I_{sc} = \frac{V}{X} = I \times \left(\frac{100}{\%X}\right)$$

↳ Advantage of percentage reactance over ohmic reactance
↳ %X reactance unchanged as they are referred through transformers
↳ Unlike ohmic reactances which are multiplied or divided by the square of transformation ratio.

Percentage Reactance and Base KVA

↳ Percentage reactance of an equipment depends upon its KVA rating
↳ Various equipments in a power system have a different KVA rating
↳ Necessary to find percentage reactance of all elements on a common KVA
↳ This is known as base KVA.
↳ The value is quite unimportant and may be,
↳ (i) Equal to that of a largest plant
↳ (ii) Equal to ~~the rated capacity~~ the total plant capacity
↳ (iii) Any arbitrary value
↳ The conversion is as follows,

$$\%\text{ reactance at base KVA} = \frac{\text{Base KVA}}{\text{Rated KVA}} \times \%\text{ reactance at rated KVA}$$

Short Circuit KVA

↳ Fictitious potential at the fault point is zero
↳ It is practice to express short circuit current in terms of short circuit KVA based on normal system voltage at the point of fault
↳ The product of normal system voltage and short ckt current at point of fault expressed in KVA
↳ Short circuit KVA.

Let, V = normal phase voltage in volts
I = Full load current in amperes at base KVA
$\%X$ = percentage reactance of the system on base KVA upto the fault point

We know, $I_{sc} = I\left(\frac{100}{\%X}\right)$

∴ Short ckt KVA for 3ϕ circuit:

$$= \frac{3VI_{sc}}{1000}$$

$$= \frac{3V}{1000} \times I\left(\frac{100}{\%X}\right)$$

$$= \frac{3VI}{1000}\left(\frac{100}{\%X}\right)$$

$$= \text{Base KVA} \times \frac{100}{\%X}$$

Reactor Control of Short Circuit Currents

↳ With expanding power system, fault level (power flows into a fault) is also rising
↳ Circuit breakers must be capable of dealing with maximum possible short-circuit currents
↳ Normally the reactance under fault condition is low
↳ Currents can rise to dangerously high levels
↳ Steps must be taken to limit the short ckt currents
↳ Duty required by circuit breakers will be excessively heavy & damage will occur to the other equipments.

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↳ In order to limit the short ckt currents additional reactances are inserted in the circuit
↳ These are called reactors (connected in series at suitable points)
↳ Reactor is a coil with a number of turns designed to have a large reactance as compared to its ohmic resistance
↳ Since the flux in these reactors is considerable so, the windings must be solidly braced
↳ Due to the very small resistance of the reactors, there is very little change in the efficiency of the system

↳ Advantages:
↳ (i) Reactors limit the flow of short ckt current
↳ protecting equipment from over heating and failure due to destructive mechanical forces
↳ (ii) Troubles are localised and isolated at the point they originate without communicating their effects to other parts
↳ This increases the chance of continuity of supply
↳ (iii) They permit the installation of ckt breakers of lower rating

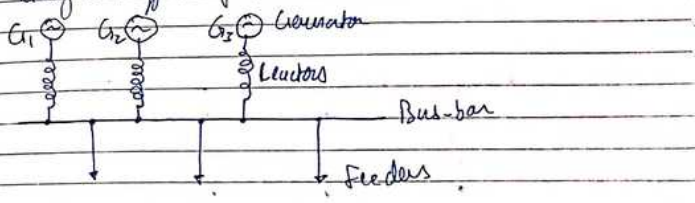
Location of Reactors

↳ Reactors may be connected in:
↳ (i) Series with each generator
↳ (ii) Series with each feeder
↳ (iii) In bus bars

↳ (i) Generator Reactors

↳ When reactors are connected in series with each generator
↳ Can be considered part of the leakage reactance of the generator
↳ The effect is to protect the generator in case of any short ckt beyond the reactors

↳ Disadvantages:
↳ (i) There is constant voltage drop and power loss in the reactors even during normal operation
↳ (ii) If a bus-bar or feeder fault occurs close to the bus-bar, the voltage of the bus bar will be reduced to a low value
↳ Generators fall out of step.
↳ (iii) If a fault occurs at any feeder, the continuity of supply is likely to be affected for the others.



↳ (ii) Feeder Reactors

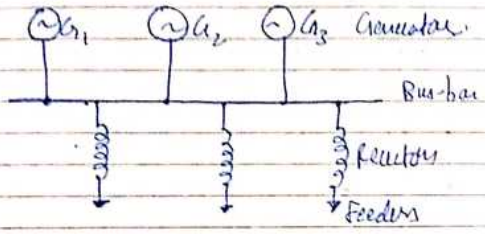
↳ When reactors are connected in series with each feeder
↳ Since most short ckt occur on feeders
↳ large no. of reactors are used

↳ Advantages:
↳ (i) If fault occurs in feeders, the voltage drop in the reactors will not affect the bus bar voltage
↳ little disadvantage for generators to lose synchronism
↳ (ii) The fault on on feeder will not affect other feeders
↳ The fault is localized

↳ Disadvantages:
↳ (i) There is a constant power loss and voltage drop even during normal operation
↳ (ii) If no short ckt occurs at a bus bar, no protection is provided

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to the generators
↳ however such faults are rare and modern generators have considerable leakage reactance
↳(iii) If the number of generators increase, the size of the feeder reactor also increases
↳ to keep the short ckt current within range of the feeder ckt breakers.

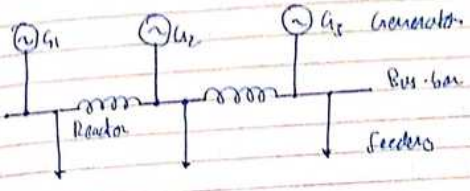


↳(iii) Bus-bar reactors

↳ The problem of voltage drop & power loss during normal operation can be overcome by using reactors in the bus-bar
↳ Two methods
↳(a) Ring System
↳(b) Tie-Bar System

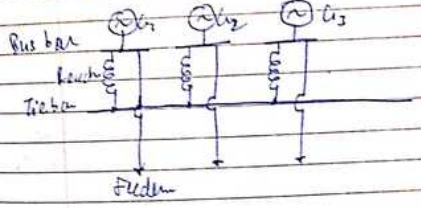
↳(a) Ring System

↳ The bus-bar is divided into sections which are connected through reactors
↳ Under normal conditions, each generator will supply its own section of the load
↳ Results in less power loss & voltage drop in reactors
↳ Under fault condition, only one section is affected



↳(ii) Tie-Bar System

↳ There are effectively two reactors in series per section
↳ Reactors must have approx. half the reactance as compared ring system
↳ Additional generators may be connected to the system without requiring changes in the reactors
↳ Disadvantage is that it requires additional bus-bar sections



Steps for Symmetrical Fault calculations

↳ Problems regarding symmetrical faults can be solved easily one phase only
↳ Procedure is as follows:
↳(i) Draw a single line diagram of the complete network
↳ indicating rating, voltage & p.u. of each element
↳(ii) Choose a convenient base kVA and convert all p.u. to this base kVA.

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↳(iii) Corresponding to the single diagram of the network, draw the reactance diagram showing one phase & the neutral
↳ Indicate the %X on the base kVA
↳ The transformer in the system should be indicated by reactance in series.
↳(iv) Find the total % reactance of the network upto the point of fault. Let it be %X.
↳(v) Find the full load current corresponding to the base kVA and the normal system voltage upto the point of fault
↳ Let it be I.
↳(vi) Then short ckt calculations are:

Short circuit current, $I_{sc} = I\left(\frac{100}{\%X}\right)$

Short ckt kVA = Base kVA $\times \frac{100}{\%X}$

Unsymmetrical Faults on 3phase system

↳ Those faults which give rise to unsymmetrical fault currents
↳ Current in the three lines becomes unequal and so does the phase displacement among them.
↳ 3 ways in which unsymmetrical fault occurs:
↳(i) Single line-to-ground fault (L-G)
↳(ii) Line to Line fault (L-L)
↳(iii) Double line to ground fault (2-L-G)



L-G fault L-L fault 2-L-G fault

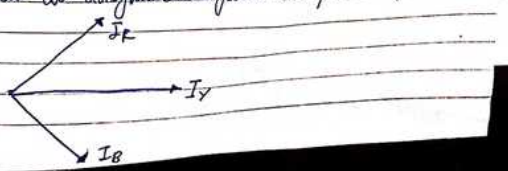
↳ Unsymmetrical fault problems can be solved by:
↳(a) Kirchoff's law
↳(b) Symmetrical components method
↳ This is preferred because
↳(a) It is a simple method and gives more generality
↳(b) It provides a useful tool for the protection engineers
↳ particularly in connection with tracing out fault currents

Symmetrical Components Method

↳ In 1918, Dr C.L. Fortescue an American scientist showed that any unbalanced system of 3ϕ currents (or be (or voltages) may be regarded as being composed of 3 separate sets of balanced vectors
↳(i) A balanced system of 3ϕ currents having positive phase sequence
↳ Positive Phase Sequence Components
↳(ii) A balanced system of 3ϕ currents having the opposite or negative phase sequence
↳ Negative Phase Sequence Components
↳(iii) A system of 3 currents equal in magnitude & having zero phase displacement
↳ Zero phase sequence components

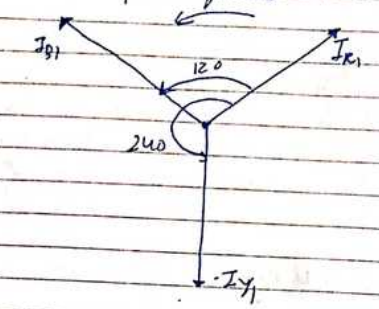
↳ The positive, negative & zero phase sequence components are called the symmetrical components of the original unbalanced system.

↳ Suppose we have an unsymmetrical fault with phase sequence RYB

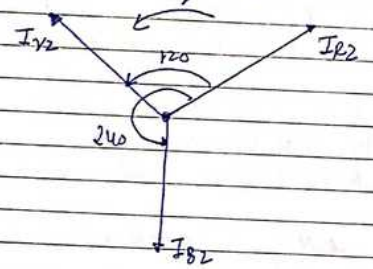


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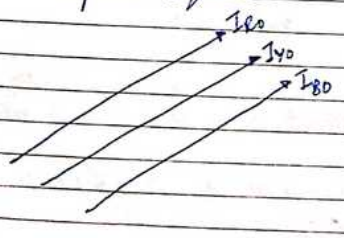
↳(i) Positive phase sequence



↳(ii) Negative Phase sequence



↳(iii) Zero phase sequence



↳ Current in any phase is the vector sum of positive, negative & zero phase sequence currents

R $\quad I_R = I_{R1} + I_{R2} + I_{R0}$

Y $\quad I_Y = I_{Y1} + I_{Y2} + I_{Y0}$

B $\quad I_B = I_{B1} + I_{B2} + I_{B0}$

$\vec{I}_R = \vec{I}_{R1} + \vec{I}_{R2} + \vec{I}_{R0}$
$\vec{I}_Y = \vec{I}_{Y1} + \vec{I}_{Y2} + \vec{I}_{Y0}$
$\vec{I}_B = \vec{I}_{B1} + \vec{I}_{B2} + \vec{I}_{B0}$

↳ Points to be noted:

↳(i) The positive, negative & zero phase sequence currents equally form balanced system of currents
↳ hence they are called symmetrical components of unbalanced system

↳(ii) The symmetrical component theory applies equally to 3φ currents & voltages both line & phase values.

↳(iii) The symmetrical components do not have a separate existance

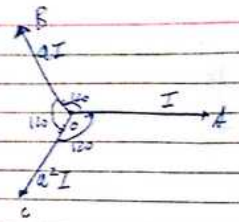
↳(iv) In a balanced 3φ system negative & zero phase sequence currents are zero.

Operator 'a'

↳ As the Symm. Comp. theory involves the concept of 120° displacement
↳ It is desirable to define an operator which would cause a 120° rotation
↳ Operator 'a' (h or λ also used)

↳ The operator 'a' is one, which when multiplied to a vector, the vector rotates through 120° in anticlockwise direction.

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$aI = I\angle 120°$

$= I(\cos 120 + j\sin 120)$

$= I(-0.5 + j0.866)$

$a = (-0.5 + j0.866)$

$a^2 I = I\angle 240°$

$= I(-0.5 - j0.866)$

$a^2 = (-0.5 - j0.866)$

$a^3 I = I$

$a^3 = 1$

↳ Properties of 'a':

↳(i) $a + a^2 = (-0.5 + j0.866) + (-0.5 - j0.866) = -1$

$\therefore 1 + a + a^2 = 0$

↳(ii) $a - a^2 = (-0.5 + j0.866) - (-0.5 - j0.866) = j1.732$

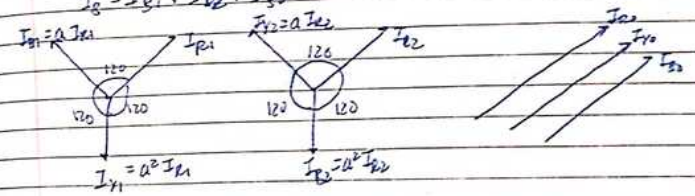
$a - a^2 = j\sqrt{3}$

Symmetrical Components in terms of phase Currents

$\vec{I}_R = \vec{I}_{R1} + \vec{I}_{R2} + \vec{I}_{R0}$

$\vec{I}_Y = \vec{I}_{Y1} + \vec{I}_{Y2} + \vec{I}_{Y0}$

$\vec{I}_B = \vec{I}_{B1} + \vec{I}_{B2} + \vec{I}_{B0}$



$\vec{I}_R = \vec{I}_{R1} + \vec{I}_{R2} + \vec{I}_{R0}$ —(i)

$\vec{I}_Y = \vec{I}_{Y1} + \vec{I}_{Y2} + \vec{I}_{Y0}$

$= a^2 I_{R1} + a I_{R2} + I_{R0}$ —(ii)

$\vec{I}_B = \vec{I}_{B1} + \vec{I}_{B2} + \vec{I}_{B0}$

$= a \vec{I}_{R1} + a^2 I_{R2} + I_{R0}$ —(iii)

↳ By adding (i)+(ii)+(iii) we get [Zero sequence current]

$\vec{I}_R + \vec{I}_Y + \vec{I}_B = \vec{I}_{R1}(1 + a^2 + a) + \vec{I}_{R2}(1 + a + a^2) + 3\vec{I}_{R0}$

[we know, $1 + a + a^2 = 0$]

$\Rightarrow \vec{I}_R + \vec{I}_Y + \vec{I}_B = \vec{I}_{R1}(0) + \vec{I}_{R2}(0) + 3\vec{I}_{R0}$

$\vec{I}_{R0} = \frac{1}{3}(\vec{I}_R + \vec{I}_Y + \vec{I}_B)$

$\Rightarrow \boxed{\vec{I}_0 = \frac{1}{3}(\vec{I}_R + \vec{I}_Y + \vec{I}_B)}$

↳ By $a \times$(ii) & a^2(iii) & (i) + (ii)a + (iii)a^2 [Positive Sequence Currents]

$\vec{I}_R + a\vec{I}_Y + a^2\vec{I}_B = \vec{I}_{R1}(1 + a^2 a^2 + a a) + \vec{I}_{R2}(1 + a^2 a + a a^2) + \vec{I}_{R0}(1 + a + a^2)$

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$= 3\vec{I}_{R1} + \vec{I}_{R2}(0) + \vec{I}_{R0}(0)$

$= 3\vec{I}_R$

$$\boxed{\vec{I}_1 = \frac{1}{3}\left(\vec{I}_R + a\vec{I}_Y + a^2\vec{I}_B\right)}$$

↳ By (ii) × a^2 & add to (i) + a^2(ii) + a(iii) [Negative Sequence (ve)]

$$\vec{I}_R + a^2\vec{I}_Y + a\vec{I}_B = \vec{I}_{R1}(1 + a^4 + a^2) + \vec{I}_{R2}(1 + a^2 + a^3) + \vec{I}_{R0}(1 + a^2 + a)$$

$$= \vec{I}_{R1}(0) + \vec{I}_{R2}(3) + \vec{I}_{R0}(0)$$

$$= 3\vec{I}_{R2}$$

$$\boxed{\vec{I}_2 = \frac{1}{3}\left(\vec{I}_R + a^2\vec{I}_Y + a\vec{I}_B\right)}$$

↳ The currents $\vec{I}_1$, $\vec{I}_2$ & $\vec{I}_0$ are symmetrical components of R-ph
↳ Due to symmetry of each set Y & B can be easily known

↳ The method applies equally to voltages,

$$\vec{E}_1 = \frac{1}{3}\left(\vec{E}_R + \vec{E}_Y + \vec{E}_B\right)$$

$$\vec{E}_2 = \frac{1}{3}\left(\vec{E}_R + a\vec{E}_Y + a^2\vec{E}_B\right)$$

$$\vec{E}_3 = \frac{1}{3}\left(\vec{E}_R + a^2\vec{E}_Y + a\vec{E}_B\right)$$

Some facts about sequence Currents

↳(i) A balanced 3ϕ system consists of +ve phase sequence components
↳ -ve & 0 sequence components are zero

↳(ii) The presence of negative & zero sequence currents in a 3ϕ system introduces unsymmetry

↳(iii) The vector sum of positive negative sequence currents of an unbalanced system is zero
↳ The resultant consists of zero sequence currents
$= \vec{I}_{R0} + \vec{I}_{Y0} + \vec{I}_{B0}$

↳(iv) In a 3ϕ 4 wire system, the magnitude of zero sequence components is 1/3 of the current in neutral wire.

$$\text{Zero sequence current} = \frac{1}{3}[\text{current in neutral wire}]$$

↳(v) In a 3ϕ unbalanced system, the magnitudes of negative sequence components cannot exceed positive sequence
↳ Phase sequence of the resultant would be reversed drawn

↳(vi) The current of a single ϕ load ~~drawn~~ from a 3ϕ system comprises equal positive, negative & zero sequence components

Sequence Impedance

↳ Each element of power system will offer different impedance to different phase sequence components of current
↳ In unsymmetrical fault calculations, each piece of equipment will have 3 values of ~~current~~ impedance,
↳(i) Positive Sequence Impedance (Z_1)
(ii) Negative Sequence Impedance (Z_2)
(iii) Zero sequence Impedance (Z_0)

↳ In a 3ϕ balanced system, each piece of equipment only offers one impedance which is positive sequence impedance
↳ The positive & negative sequence impedance of linear, symmetrical & static circuits are the same and the same as balanced condition
↳ like transmission lines, cables, transformers & static loads &
↳ Impedance of such ckt is independent to phase orders.

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↳ Zero sequence impedance depends upon the path taken by the zero sequence current
↳ As this path is generally different from positive & negative sequence currents
↳ Zero sequence impedance is generally different from positive & negative sequence impedance

Sequence Impedance of Power System Elements

↳ (i) Synchronous Generators

↳ The positive & negative sequence impedance of rotating machines are generally different
↳ The positive sequence impedance of a synchronous generator is called as synchronous impedance
↳ The negative sequence impedance is usually low
↳ The zero sequence impedance is a variable item
↳ Usually taken as equal to the positive sequence impedance

↳ (ii) Transformers

↳ Since transformers have the same impedance with reversed phase rotation
↳ Positive and negative sequence impedances are equal
↳ Being equal to the impedance of the transformer
~~↳ Zero sequence impedance is usually greater than the positive sequence impedance~~

Positive sequence impedance = Negative sequence impedance
= Impedance of transformer
Zero Sequence impedance = Positive seq. imp. if Neut's circuit for earth current
= infinite, if not

↳ (iii) Transmission lines

↳ Positive sequence impedance = Negative sequence impedance
= Impedance of the line

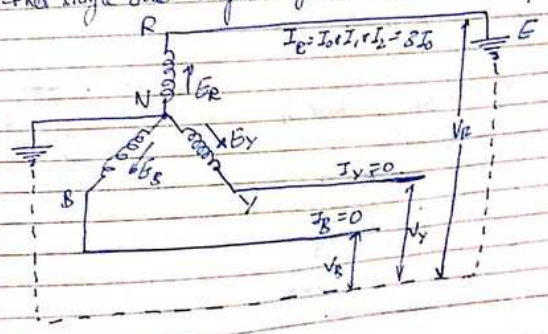
↳ Zero sequence impedance = Variable item
= may be taken as 5x positive seq. imp.

Analysis of Unsymmetrical faults

↳ Following assumptions will be made
↳ (i) The generated emf system is of positive sequence only
↳ (ii) No current flows in the system other than due to faults
↳ Load current is neglected
↳ (iii) The impedance of the fault is zero
↳ (iv) Phase R shall be taken as the reference phase.

↳ Single line-to-Ground Fault

↳ Consider a 3ϕ system with an earthed neutral
↳ Let single line-to-ground fault occur in the Red phase:



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Now, $\vec{V}_c = 0$

$\vec{I}_B = \vec{I}_y = 0$

The sequence currents in the red phase in terms of line currents:

$$\vec{I}_0 = \frac{1}{3}(\vec{I}_R + \vec{I}_y + \vec{I}_B) = \frac{1}{3}\vec{I}_R$$

$$\vec{I}_1 = \frac{1}{3}(\vec{I}_R + a\vec{I}_y + a^2\vec{I}_B) = \frac{1}{3}\vec{I}_R$$

$$\vec{I}_2 = \frac{1}{3}(\vec{I}_R + a^2\vec{I}_y + a\vec{I}_B) = \frac{1}{3}\vec{I}_R$$

$$\vec{I}_0 = \vec{I}_1 = \vec{I}_2 = \frac{1}{3}\vec{I}_R$$

↳ **Fault currents:**

↳ Let $\vec{Z}_0, \vec{Z}_1, \vec{Z}_2$ be the zero, positive & negative sequence impedances of the generator respectively

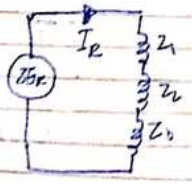
↳ Consider the closed loop RBGN:

$$\vec{E}_R = \vec{I}_1\vec{Z}_1 + \vec{I}_2\vec{Z}_2 + \vec{I}_0\vec{Z}_0 + \vec{V}_R$$

But, $\vec{V}_R = 0$ & $\vec{I}_1 = \vec{I}_2 = \vec{I}_0$

$$\vec{E}_R = \vec{I}_0(\vec{Z}_0 + \vec{Z}_1 + \vec{Z}_2)$$

$$\vec{I}_0 = \frac{\vec{E}_R}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0} \quad \text{but } \vec{I}_R = 3\vec{I}_0 \Rightarrow \boxed{\vec{I}_R = \frac{3\vec{E}_R}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}}$$ [Fault current]



If the fault impedance is Z_f instead of zero

$$\vec{I}_R = \frac{3\vec{E}_R}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0 + 3\vec{Z}_f}$$

↳ If the neutral is not grounded

↳ No path for fault current

↳ Zero sequence impedance will be infinite

↳ **Phase Voltage at Faults:**

↳ Generated power system is of positive sequence only

↳ Sequence components of fault in R phase:

$$\vec{E}_0 = 0; \quad \vec{E}_2 = 0; \quad \vec{E}_1 = \vec{E}_R$$

↳ Sequence voltages at fault for Rϕ:

$$\vec{V}_1 = \vec{E}_R - \vec{I}_1\vec{Z}_1$$

$$\left[I_1 = I_0 = \frac{\vec{E}_R}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}\right]$$

$$\Rightarrow \vec{V}_1 = \vec{E}_R - \frac{\vec{E}_R\vec{Z}_1}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}$$

$$\boxed{\vec{V}_1 = \left(\frac{\vec{Z}_2 + \vec{Z}_0}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}\right)\vec{E}_R}$$

$\vec{V}_1 = \vec{E}_R + \vec{I}_1\vec{Z}_1 + \vec{I}_2\vec{Z}_2 + \vec{I}_0\vec{Z}_0$

$0 = \vec{I}_2\vec{Z}_2 + \vec{I}_0\vec{Z}_0 + \vec{V}_2$

& $\vec{V}_2 = 0 - \vec{I}_2\vec{Z}_2$

$$\boxed{\vec{V}_2 = \left(\frac{-\vec{Z}_2}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}\right)\vec{E}_R}$$

$$\boxed{\vec{V}_0 = \left(\frac{-\vec{Z}_0}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}\right)\vec{E}_R}$$

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↳ Clearly $\vec{V}_1 + \vec{V}_2 + \vec{V}_0 = 0$

Phase Voltages are as follows :-

$\vec{V}_R = \vec{V}_0 + \vec{V}_1 + \vec{V}_2 = 0$

$\vec{V}_Y = \vec{V}_0 + a\vec{V}_1 + a^2\vec{V}_2$

$\vec{V}_B = \vec{V}_0 + a\vec{V}_1 + a^2\vec{V}_2$

↳ Summary of Results :

↳ $\vec{I}_f$ = Fault Current $= \dfrac{3\vec{E}_a}{\vec{Z}_1 + \vec{Z}_2 + \vec{Z}_0}$

↳ $\vec{I}_Y = \vec{I}_B = 0$

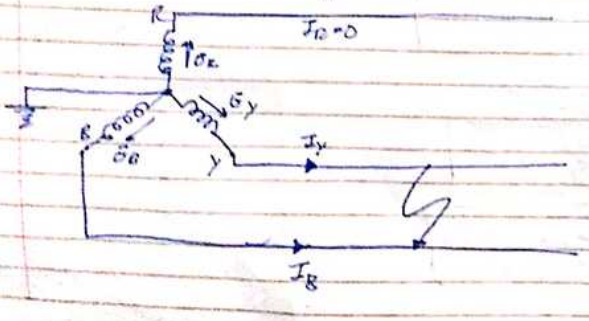
↳ $\vec{V}_R = 0$

↳ $\vec{V}_Y = \vec{V}_0 + a^2\vec{V}_1 + a\vec{V}_2$

↳ $\vec{V}_B = \vec{V}_0 + a\vec{V}_1 + a^2\vec{V}_2$

↳ Line-to-line Fault

↳ Consider a line-to-line fault b/w B & Y



$\vec{V}_Y = \vec{V}_B$

$\vec{I}_R = 0$

$\vec{I}_Y + \vec{I}_B = 0$

Taking R-phase as reference,

$\vec{I}_0 = \frac{1}{3}(\vec{I}_R + \vec{I}_B + \vec{I}_Y) = 0$

Now, $\vec{V}_Y = \vec{V}_B$

$\Rightarrow V_0 + a^2\vec{V}_1 + a\vec{V}_2 = V_0 + a\vec{V}_1 + a^2\vec{V}_2$

$\vec{V}_1(a^2 - a) = V_2(a^2 - a)$

$\boxed{\vec{V}_1 = \vec{V}_2}$ —(i)

Also, $\vec{I}_Y + \vec{I}_B = 0$

$\Rightarrow \vec{I}_0 + a^2\vec{I}_1 + a\vec{I}_2 + \vec{I}_0 + a\vec{I}_1 + a^2\vec{I}_2 = 0$

$2\vec{I}_0 + (a^2 + a)\vec{I}_1 + (a^2 + a)\vec{I}_2 = 0$

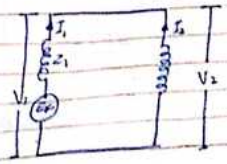
$2\vec{I}_0 + (a^2 + a)(\vec{I}_1 + \vec{I}_2) = 0$ $[I_0 = 0]$

$\boxed{\vec{I}_1 + \vec{I}_2 = 0}$ —(ii)

↳ Fault Currents:

$\vec{I}_1 = -\vec{I}_2$

$= \dfrac{E_a}{\vec{Z}_1 + \vec{Z}_2}$



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Fault current, $I_y = \vec{I}_0 + a^2\vec{I}_1 + a\vec{I}_2$

$= 0 + a^2\left(\frac{\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}\right) + a\left(\frac{-\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}\right)$

$= (a^2 - a)\left(\frac{\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}\right)$

$I_y = \frac{-j\sqrt{3}\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}$ $\quad [\because a^2 - a = -j\sqrt{3}]$

But $I_y = -I_B$

$\Rightarrow \vec{I}_B = \frac{j\sqrt{3}\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}$

Phase voltages

↳ Since, the generated phase sequence is of positive phase sequence only:

$\vec{E}_0 = 0$; $\vec{E}_2 = 0$ and $\vec{E}_1 = \vec{E}_R$

$\vec{V}_1 = \vec{E}_R - \vec{I}_1\vec{Z}_1 = \vec{E}_R - \frac{\vec{E}_R\vec{Z}_1}{\vec{Z}_1+\vec{Z}_2}$

$$\boxed{\vec{V}_1 = \left(\frac{\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_R}$$

$\vec{V}_2 = 0 - \vec{I}_2\vec{Z}_2 = \left(\frac{\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_R$

$$\boxed{\vec{V}_2 = \left(\frac{\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_A}$$

$\vec{V}_0 = 0 - \vec{I}_0 Z_0$

$= 0$

$\vec{V}_R = \vec{V}_1 + \vec{V}_2 + \vec{V}_0$

$= \left(\frac{\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_R + \left(\frac{\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_A + 0$

$\vec{V}_R = \left(\frac{2\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_R$

$\vec{V}_Y = \vec{V}_0 + a^2\vec{V}_1 + a\vec{V}_2$

$= 0 + (a^2 + a)\left(\frac{\vec{Z}_2\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}\right)$

$\vec{V}_Y = -\frac{\vec{Z}_2\vec{E}_A}{\vec{Z}_1+\vec{Z}_2}$

$\vec{V}_B = \vec{V}_0 + a\vec{V}_1 + a^2\vec{V}_2$

$= 0 + (a + a^2)\left(\frac{\vec{Z}_2\vec{E}_R}{\vec{Z}_1+\vec{Z}_2}\right)$

$\vec{V}_B = \left(\frac{-\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_R$

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Summary of results: Line to line fault (Blue-Yellow line)

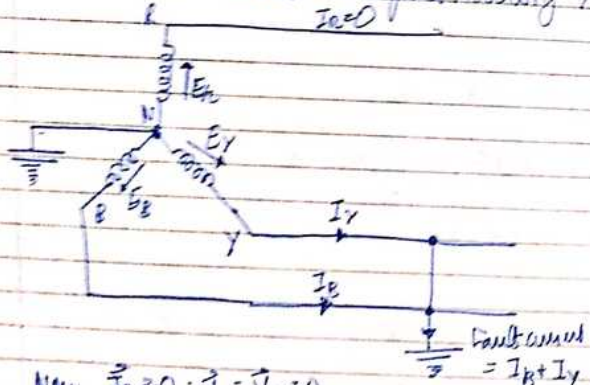
(i) $\vec{I}_B = 0$; $\vec{I}_Y = -\vec{I}_B = \frac{-j\sqrt{3}\,\vec{E}_a}{\vec{Z}_1 + \vec{Z}_2}$

(ii) $\vec{V}_2 = \vec{V}_B = \frac{-\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\vec{E}_a$

$\vec{V}_0 = \left(\frac{\vec{Z}_2}{\vec{Z}_1+\vec{Z}_2}\right)\vec{E}_a$

Double line-to-Ground Faults

Consider a double line to ground fault involving Y-B line to earth.

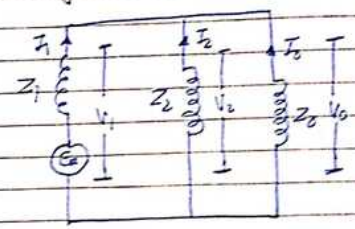


Now $\vec{I}_R = 0$; $\vec{V}_1 = \vec{V}_B = 0$

$\Rightarrow \vec{V}_1 = \vec{V}_2 = \vec{V}_0 = \frac{1}{3}\vec{V}_R$

$\vec{I}_R = \vec{I}_1 + \vec{I}_2 + \vec{I}_0 = 0$

Fault current:



$\vec{I}_1 = \frac{\vec{E}_a}{\vec{Z}_1 + \frac{\vec{Z}_2\vec{Z}_0}{\vec{Z}_2+\vec{Z}_0}}$

$\vec{I}_2 = -\vec{I}_1\frac{\vec{Z}_0}{\vec{Z}_2+\vec{Z}_0}$

$\vec{I}_0 = -\vec{I}_1\frac{\vec{Z}_2}{\vec{Z}_2+\vec{Z}_0}$

Fault current $I_f = I_Y + I_B = 3\vec{I}_0$

$I_f = 3\left(-\vec{I}_1\frac{\vec{Z}_2}{\vec{Z}_2+\vec{Z}_0}\right)$

$= 3\left(-\frac{\vec{E}_a\vec{Z}_2}{\vec{Z}_2+\vec{Z}_0}\right)\left(\frac{1}{\vec{Z}_1+\frac{\vec{Z}_2\vec{Z}_0}{\vec{Z}_2+\vec{Z}_0}}\right)$

$= 3\left(\frac{\vec{E}_a\vec{Z}_2}{\vec{Z}_2+\vec{Z}_0}\right)\left(\frac{\vec{Z}_2+\vec{Z}_0}{\vec{Z}_2\vec{Z}_1+\vec{Z}_1\vec{Z}_0+\vec{Z}_2\vec{Z}_0}\right)$

$$\boxed{I_f = \frac{-3\vec{E}_a\vec{Z}_2}{\vec{Z}_2\vec{Z}_1+\vec{Z}_1\vec{Z}_0+\vec{Z}_2\vec{Z}_0}}$$

$$\left[\vec{I}_0 = \frac{1}{3}(\vec{I}_R+\vec{I}_B+\vec{I}_Y)\right.$$
$$\left.3I_0 = (\text{fault current})\right]$$

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→ Phase Velocity:
→ Sequence Voltages in plane F:

$$\vec{V}_1 = \vec{E}_a - \vec{I}_1 Z_1 \; ; \; \vec{V}_2 = 0 - \vec{I}_2 Z_2 \; ; \; V_0 = 0 - \vec{I}_0 Z_0$$

$$\vec{V}_1 = \vec{V}_2 = \vec{V}_0 = \vec{I}_a$$

$$\vec{V}_a = \vec{V}_1 + \vec{V}_2 + V_0 = 3\vec{V}_a$$

$$\vec{V}_y = a^2 \vec{V}_1 + a \vec{V}_2 + \vec{V}_0 = (a^2 + a + 1)\vec{V}_a$$

$$\vec{V}_y = 0$$

$$\vec{V}_a = \vec{V}_y = 0$$

Sequence Networks

→ Sequence networks of a particular sequence current in the path of the flow of that sequence current in the system.
→ It is composed of interconnected diagram to find sequence current.
→ Since there are 3 sequence current, there are 3 sequence networks
→ (i) Positive Sequence Network
→ (ii) Negative Sequence Network
→ (iii) Zero Sequence Network

(i) Positive Sequence Network

→ Network all the paths for the flow of positive sequence currents
→ Following rules are used:-
(a) Each element is represented by the equivalent voltage in series with the impedance
(b) Each element is represented by its positive sequence impedance
(c) Current through the equipment is equal to the positive sequence current
(d) All protective & measuring circuits are excluded

(e) For symmetrical circuit, the shunt capacitance & resistance are neglected
(f) Only the positive sequence voltages are considered

(ii) Negative Sequence Network

→ Network all the paths for the flow of negative sequence currents
→ Following rules are used:-
(a) Each element is represented by its negative sequence impedance
(b) Current through the equipment is equal to the negative sequence current
(c) Only the negative sequence voltages are considered

(iii) Zero Sequence Network

→ Network all the paths for the flow of zero sequence currents
→ Following rules are used:-
(a) Each element is represented by its zero sequence impedance
(b) Current through the equipment is equal to the zero sequence current
(c) Only the zero sequence voltages are considered

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Reference Bus for Sequence Networks

- While drawing sequence networks, necessary to specify the reference potential w.r.t which all sequence voltage drops to be taken
  - (i) For positive or negative sequence networks
    - Neutral of the generator is taken as the reference bus
  - (ii) For zero sequence network, the reference bus is the ground at the generator

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